Week 1: overview / review

NRSC 7657 Workshop in Advanced Programming for Neuroscientists

Course outline

- Language of choice (python or MATLAB pref.)
- Didactic and practical
  - Language specifics
  - General concepts in computing (version control, debugging, unit testing, scaling)
- Independent project
  - Work with real data; preferably your own (whatever it is!) or choose a public data set (see next slide)
  - Develop an idea; ****Schedule a meeting with Dan this week****

    https://zcal.co/i/TnPlYgEZ

June 9	Week 1 – overview / review	Course overview: theory of computing, landscape of computing options. Basic usage in python and MATLAB; basic data types; environments Style guidelines (ten simple rules); git and version control
June 16	Week 2 – language fundamentals	Functions; Objects and Classes; Workspaces Typical data formats: working with tabular data, images, and time series. NeurodataWithoutBorders format
June 23	Week 3 – workflow management and outputs	Importing and exporting Packaging
June 30	Week 4 – usability	Plotting and visualization - from bar charts to 3D animation
July 7	Week 5 – scaling	Troubleshooting and debugging; unit testing
July 14	Week 6 – collaboration	Iteration and code profiling; parallel computing. Code quality-of-life topics
July 21	Week 7 – applications/flex topic	Cloud-based tools: AWS, GCC, Colab, jupyterhub, deepnote. Overview of some available SAAS tools, python focused.
		Group programming time
July 28	Week 8 – Open topics	Make-up session – open topics
August 4	Week 9 – Final presentations and code review	Final pres. and code review
		Final pres. and code review
August 11	Week 10 – Final presentations and code review	Final pres. and code review
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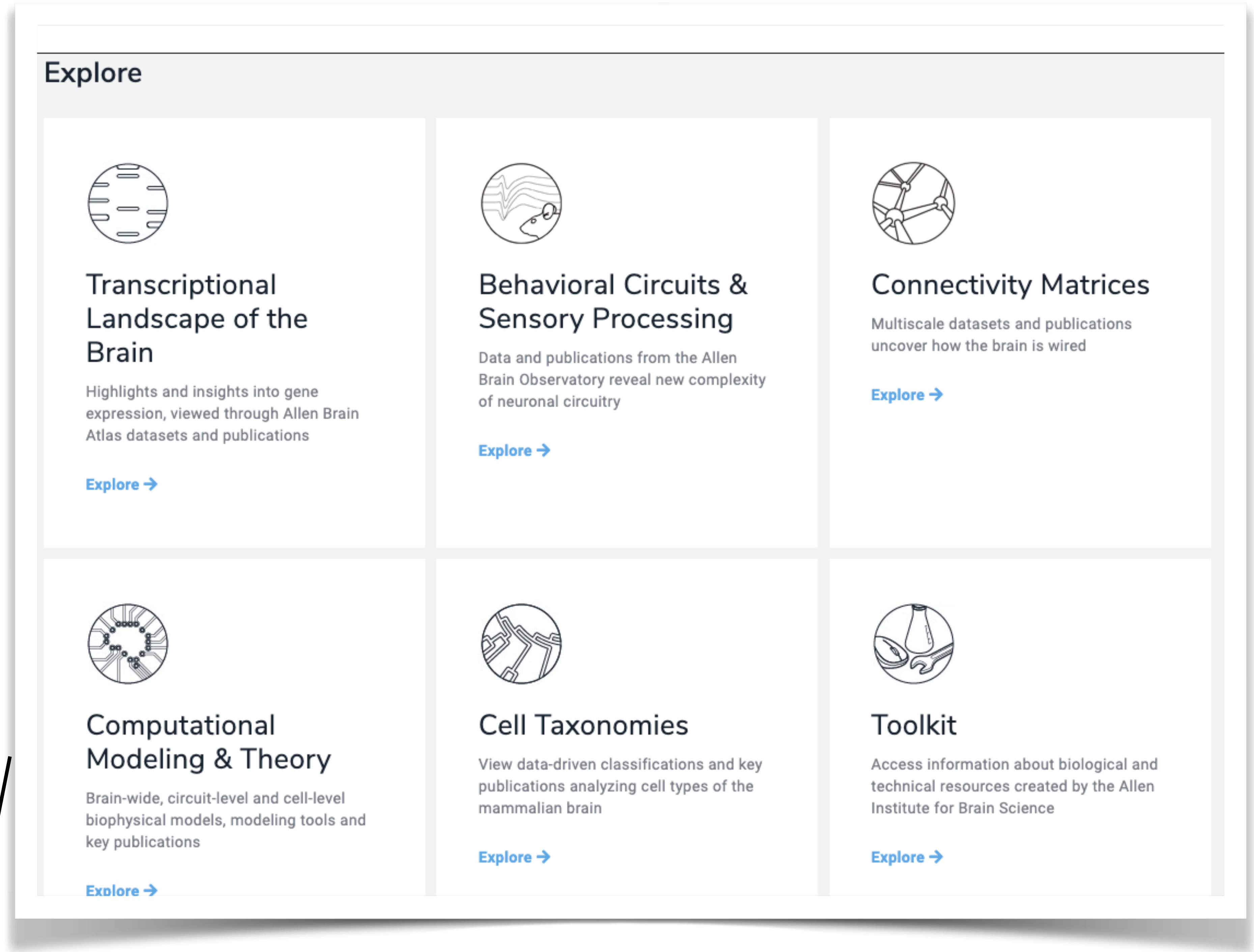
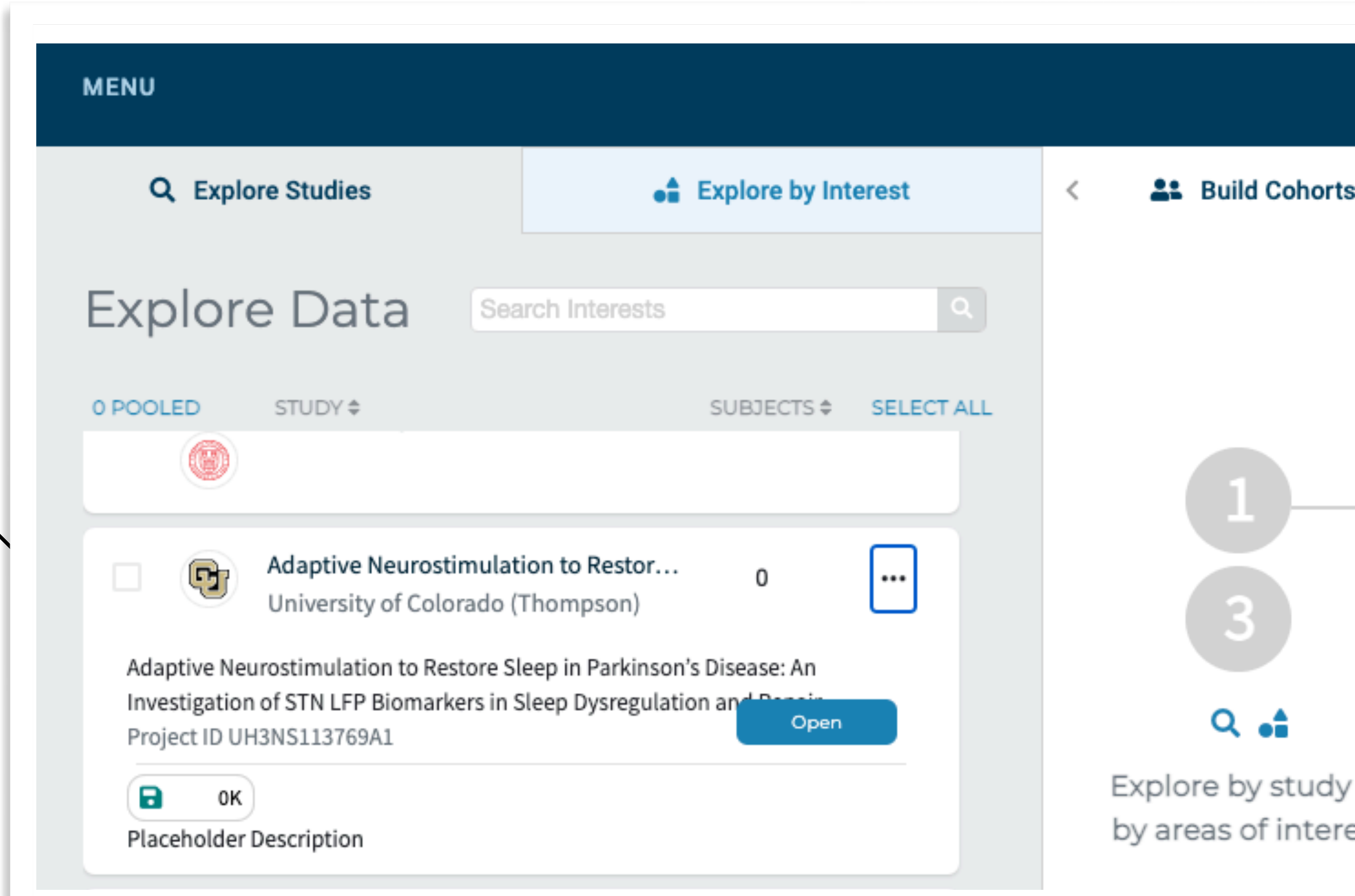
Project datasets

- Yours!
- Someone from your lab

Public datasets
(Suggestions, incomplete list)

- Allen Institute for Brain Science brain-map.org
- BRAIN Initiative archives
  - DABI https://dabi.loni.usc.edu/home
  - DANDI, https://dandiarchive.org/





What this course is **not**

- Computational neuroscience
- Mathematics
- Computer science

What this course is

Each session:

- Some slides (0 - 60 minutes), a notebook on your own time

- Working through the notebook or interactive coding together. “Flippped” classroom each week (after this week). So what we do depends on you.

- Independent coding. We’re going to do your science, with a computer. Maybe your computer, or maybe a cloud computer. Definitely your science.

content for each week available Tuesday AM for the next Monday from the course github

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- Exposure to differing approaches to computing in neuroscience
- Develop confidence in independent coding skills
- Complete a project using data relevant to your thesis work

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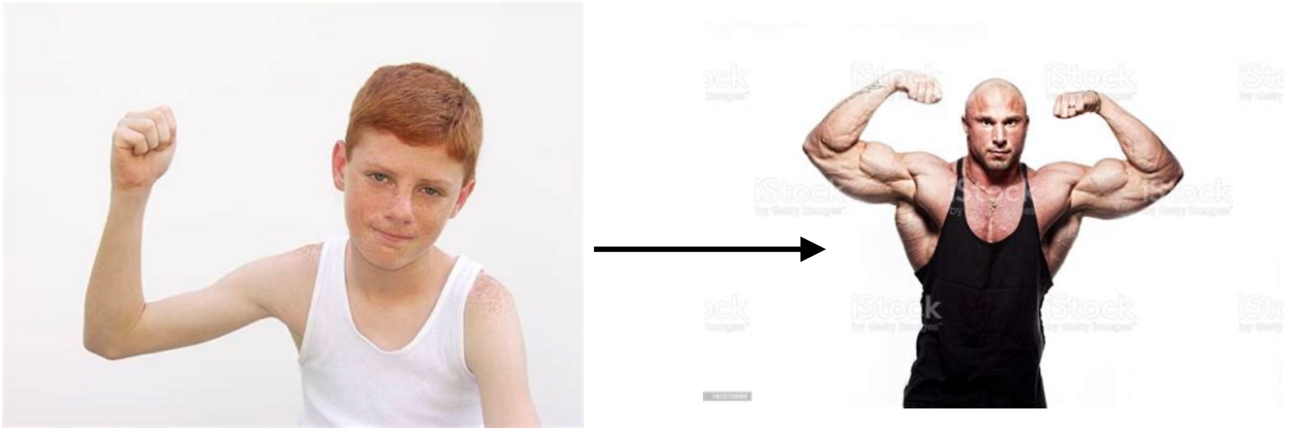
- Develop confidence in independent coding skills 
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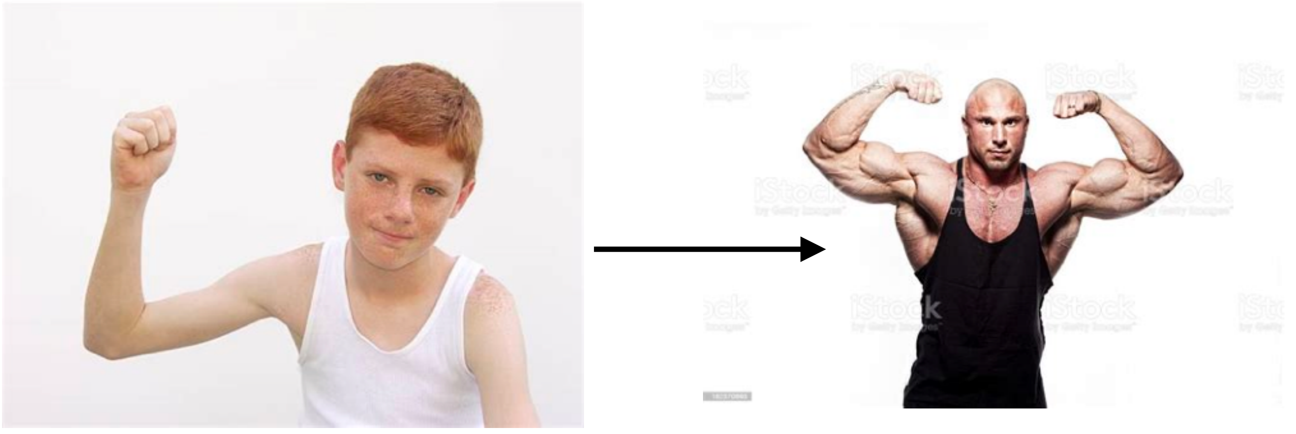
- Develop confidence in independent coding skills 
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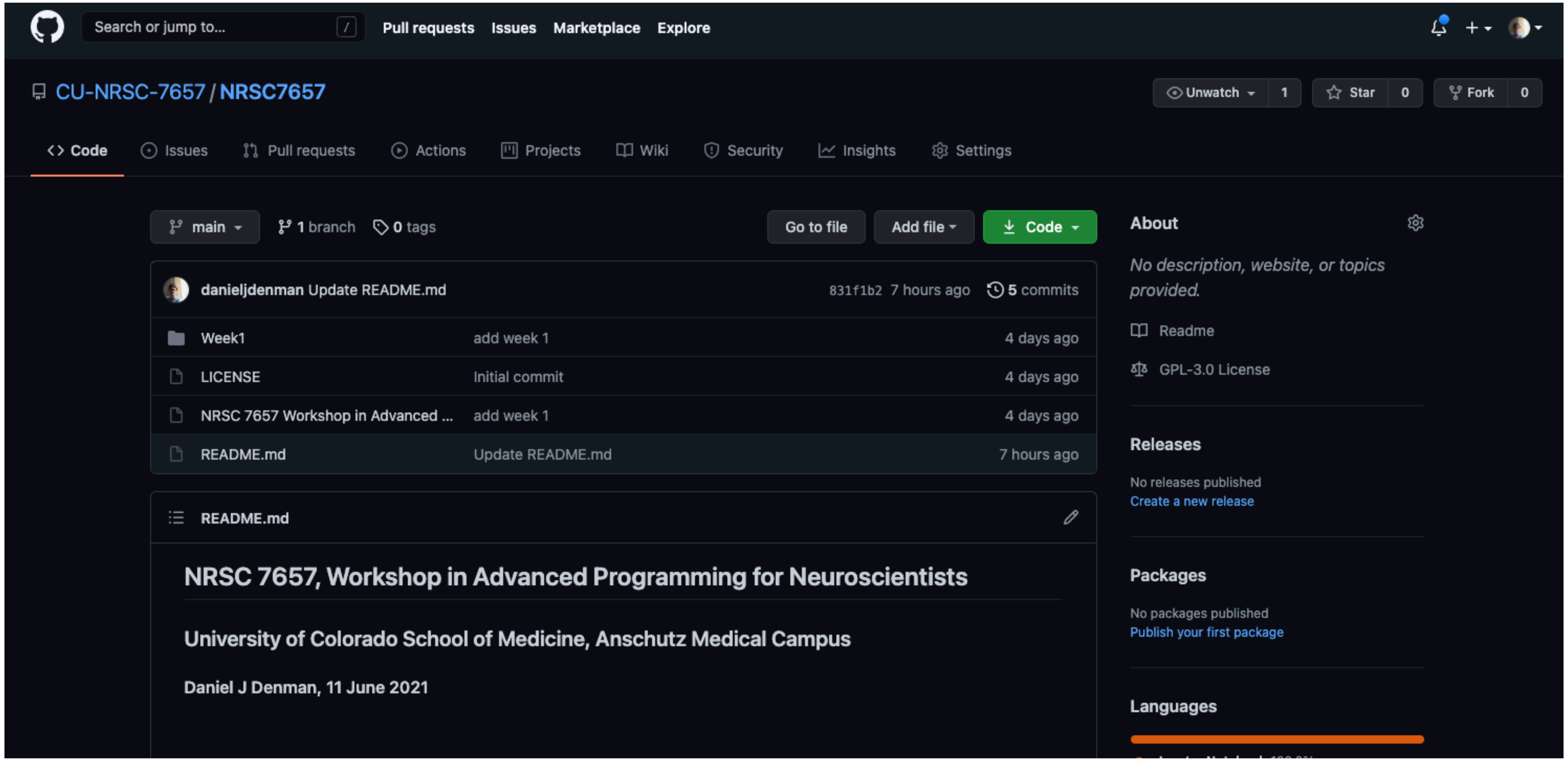
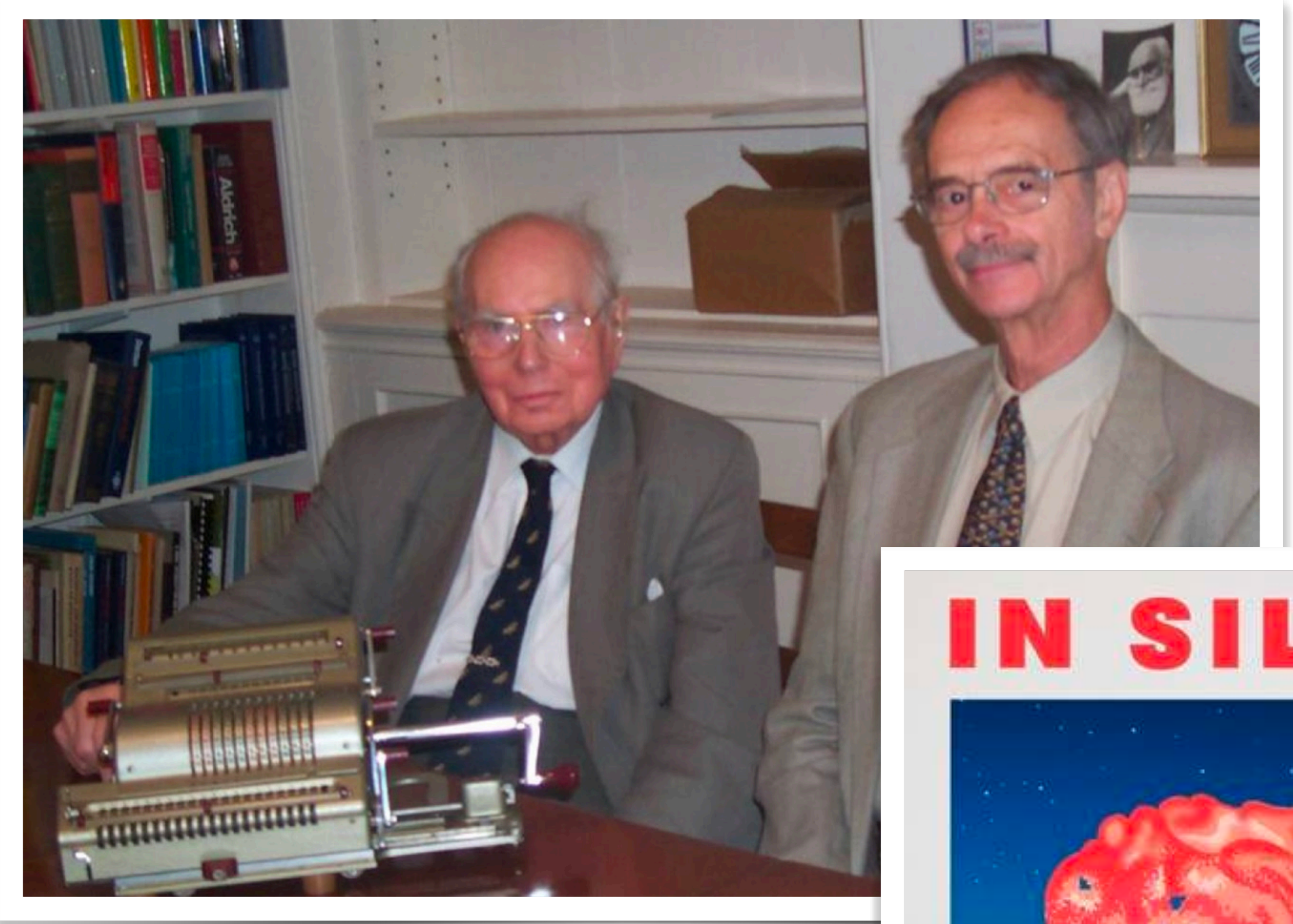
Final practicalities

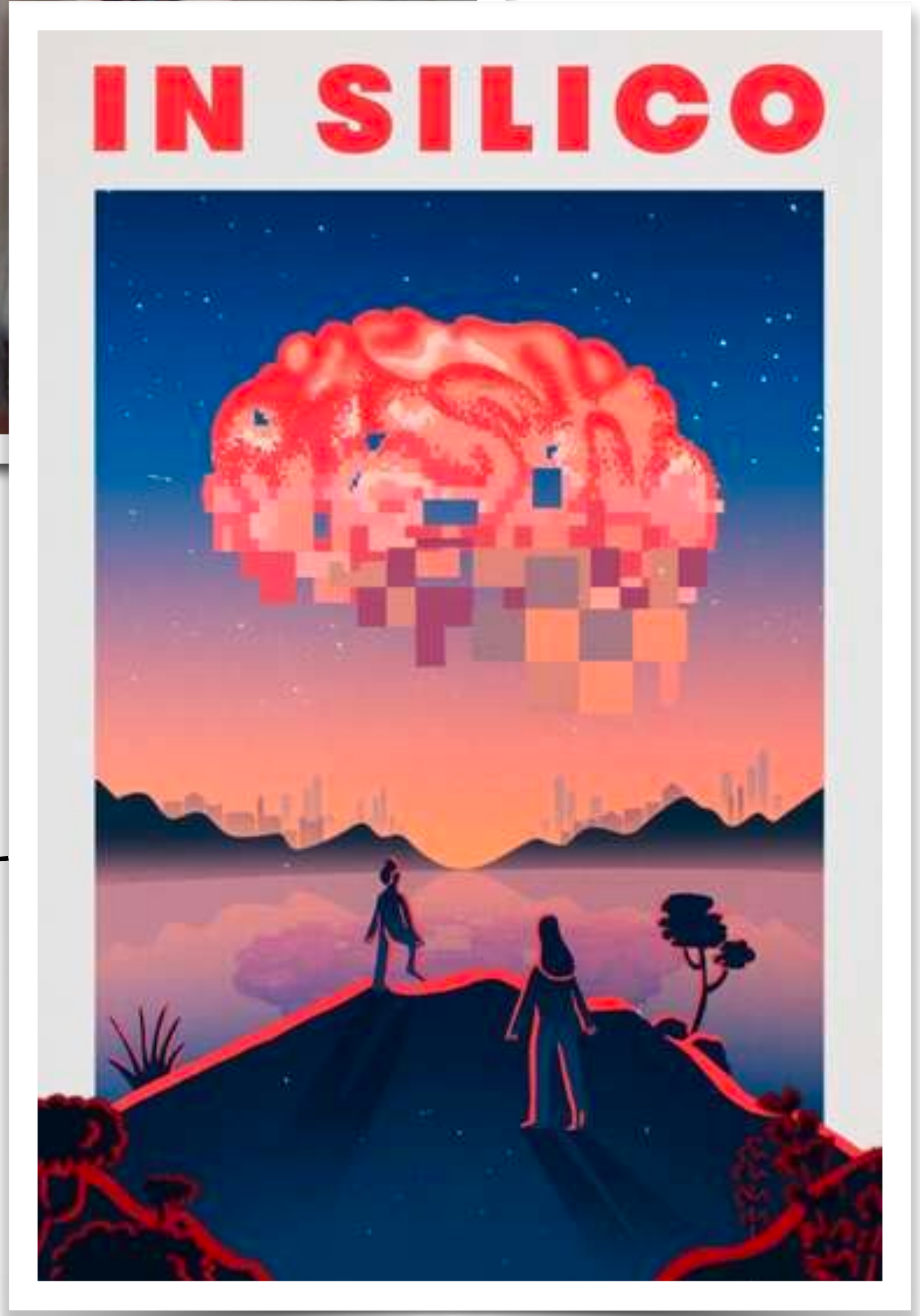
- GitHub: https://github.com/CU-NRSC-7657/NRSC7657



Computing in neuroscience

- Hodkin-Huxley's mechanical computer
- Computer based analysis / replacing chart measurements
- NEURON simulations; single neuron modeling
- …
- "data science", open data
- Blue brain project, *In silico*





Computing in neuroscience

An example: code written by neuroscientists for one experiment

planning

Probe trajectory: MATLAB

Model prediction: python - jupyter notebook

experimental control

Acquisition hardware: FPGA gate array programming

Acquisition software: C++; python plugin; [Julia]

Visual stimuli: python script; embedded python in React

Video monitoring: python script

"pre-processing"

Spike sorting: MATLAB

Unit quality: python - jupyter notebook

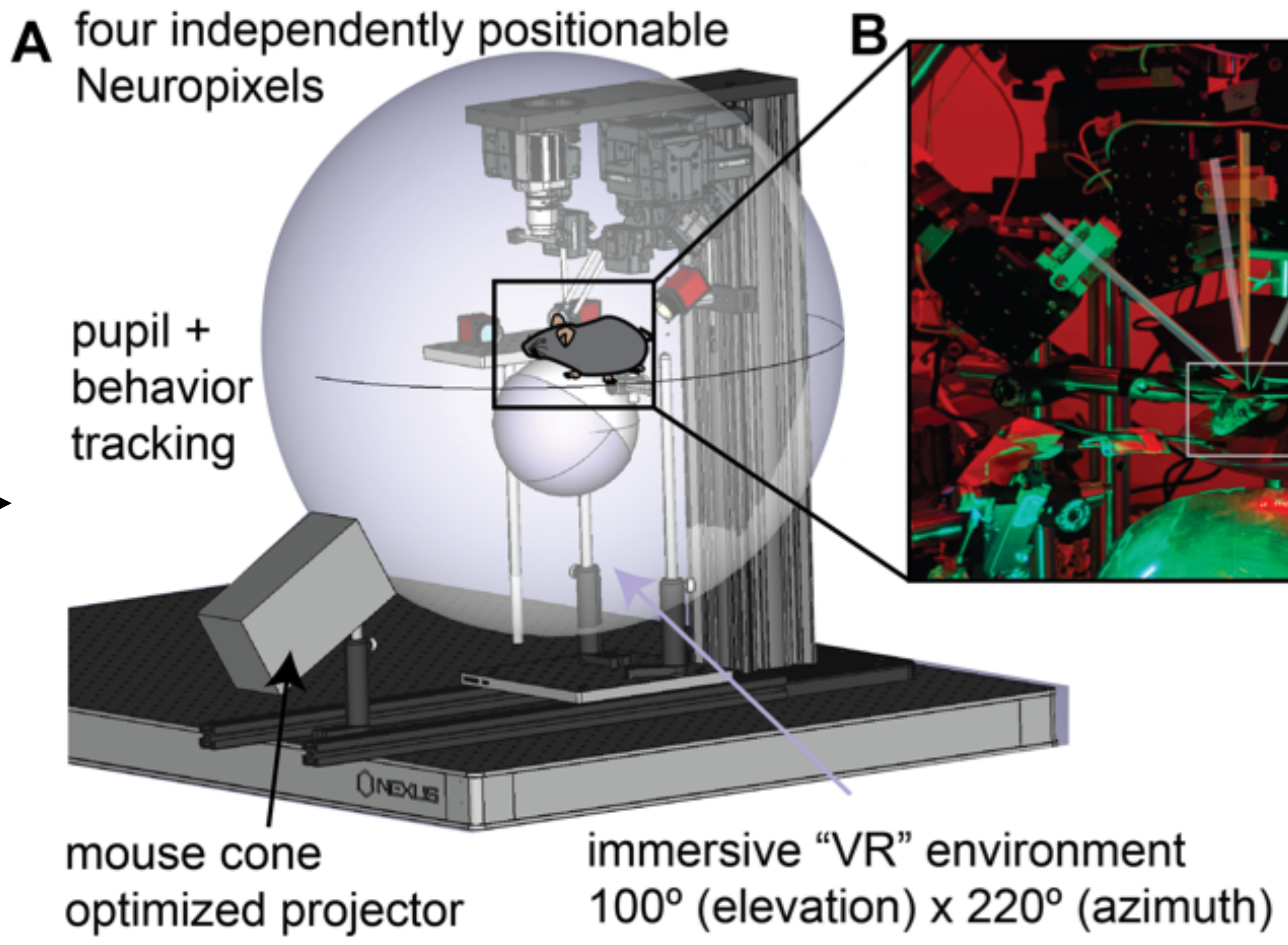
Depth: python - jupyter notebook

analysis

Histology registration to 3D brain: python

Stimulus responses: python

Population statistics: MATLAB *and* python



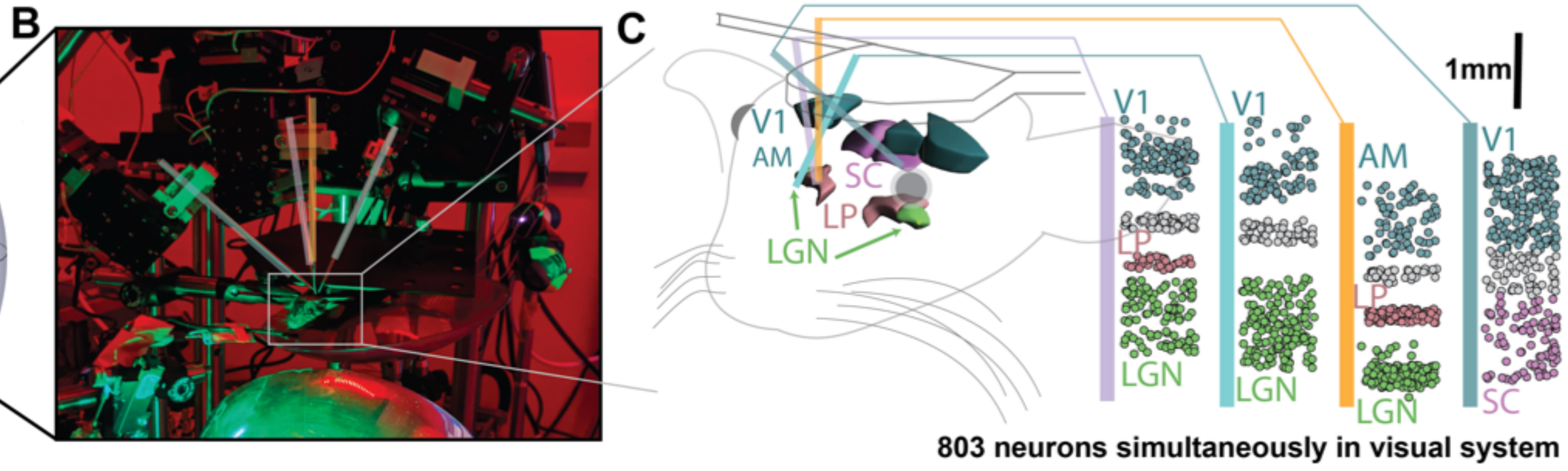


Figure 1. Platform for studying distributed visual coding. **A**., Immersive visual environment for mouse psychophysics. **B.,** Simultaneous Neuropixels recordings in the mouse visual system, spanning V1, cortical area AL, LGN thalamus, LP thalamus, and superior colliculus. **C.,** 803 single neurons isolated from the above brain areas; anatomy renderings from the Allen Reference Atlas (brain-map.org).

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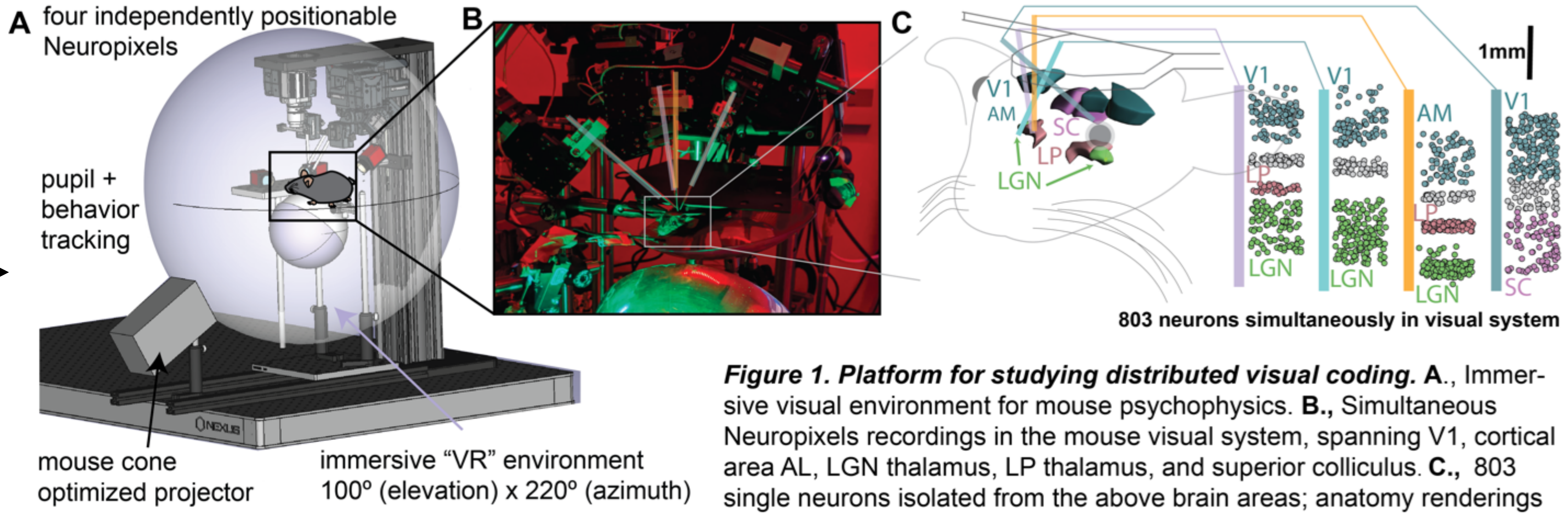

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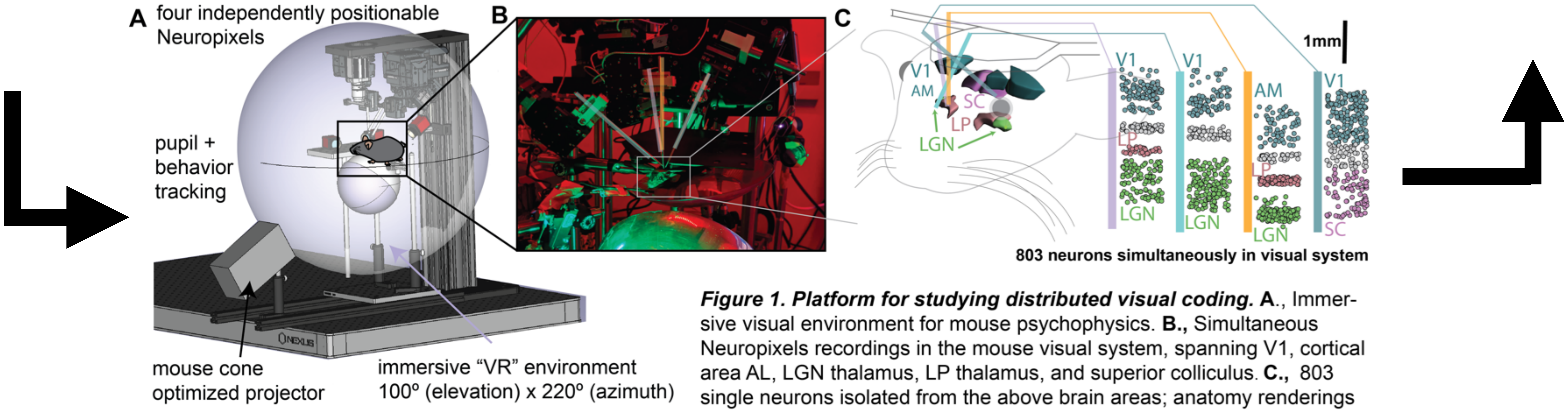

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overview

Some plusses

Some minuses

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- Free
- Readable syntax
- Cross platform
- Huge community
- Used across science **_<u>and</u>_** outside of science

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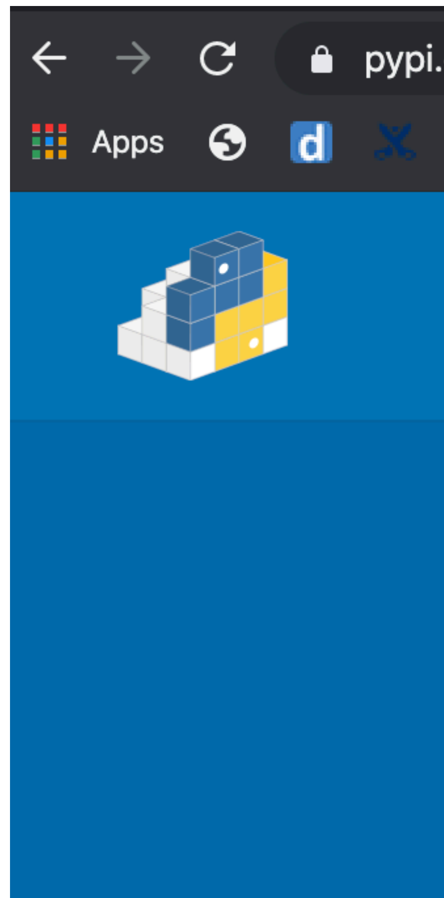
overview: packages

Packages

In addition to the standard library, the true power of python is the extensive world of packages available. These are sets of tools you can use with Python to do just about anything!

Some are general tools, the hammers or screwdrivers of using python for science: **numpy**, **matplotlib, pandas, seaborn**

Others are specialized: **scikit-learn, PIL, scanpy, Suite2P, DeepLabCut, PyTom**



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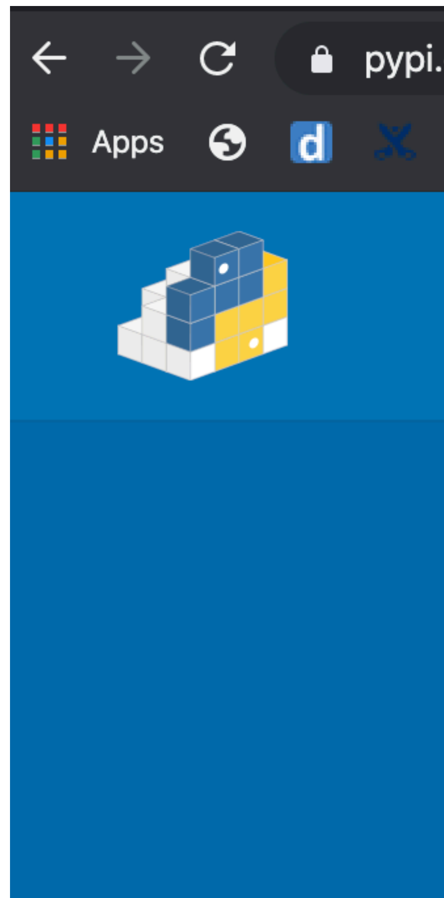

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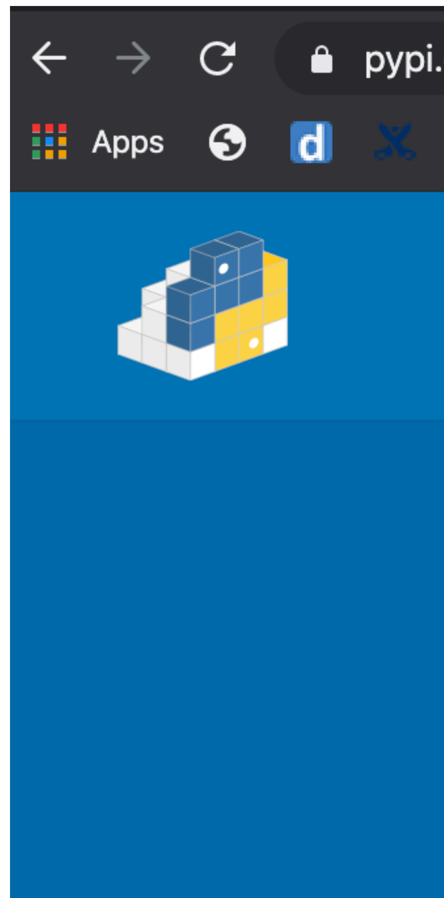

overview: levels

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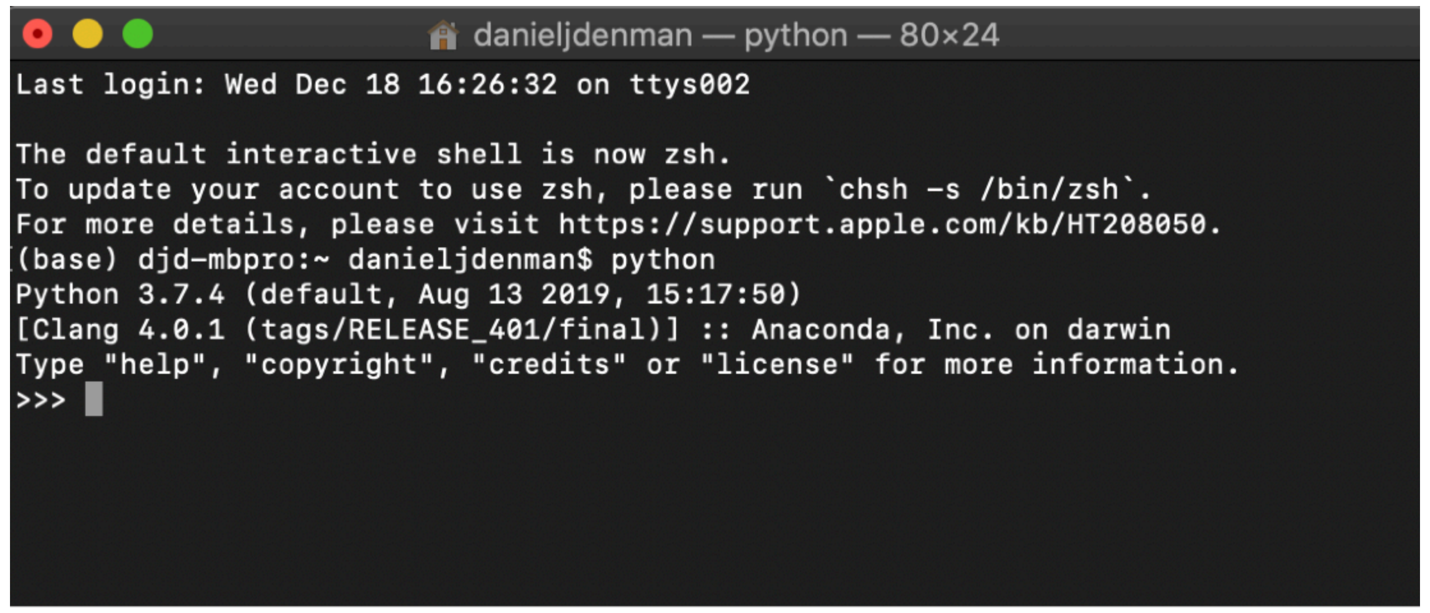

Native in Mac OS X, Linux; in Windows store (free)

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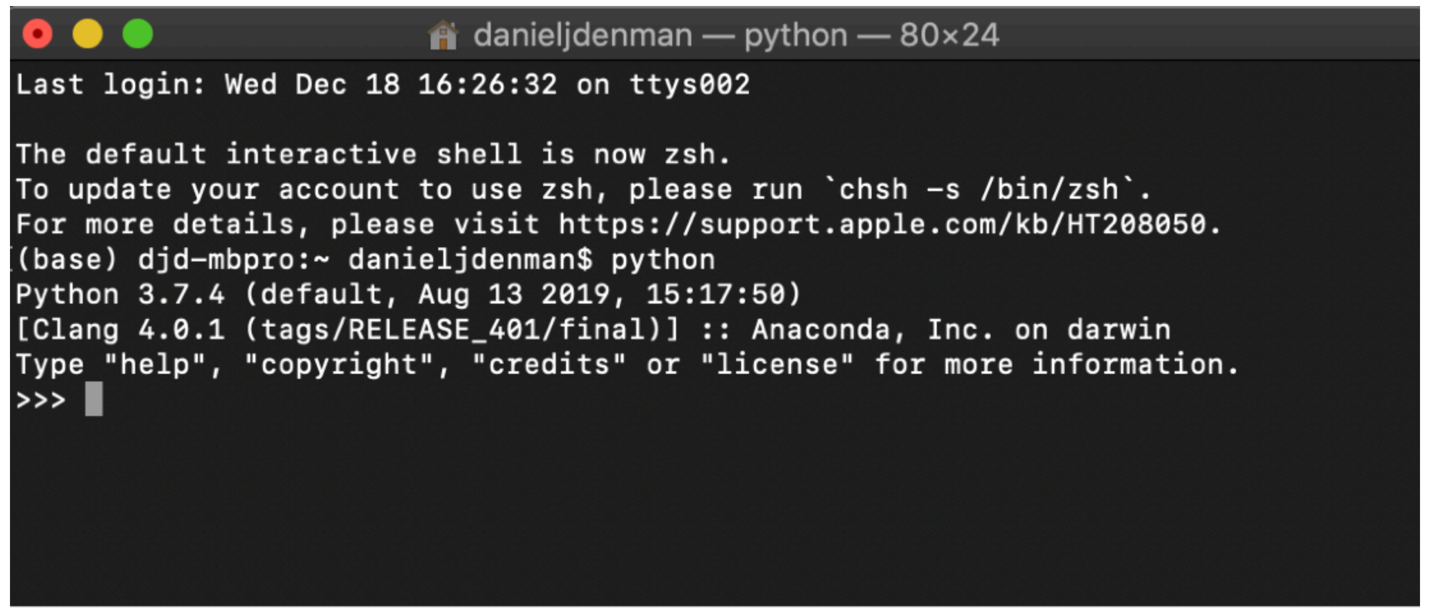

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Package Managers

Environments

ANACONDA

overview: levels

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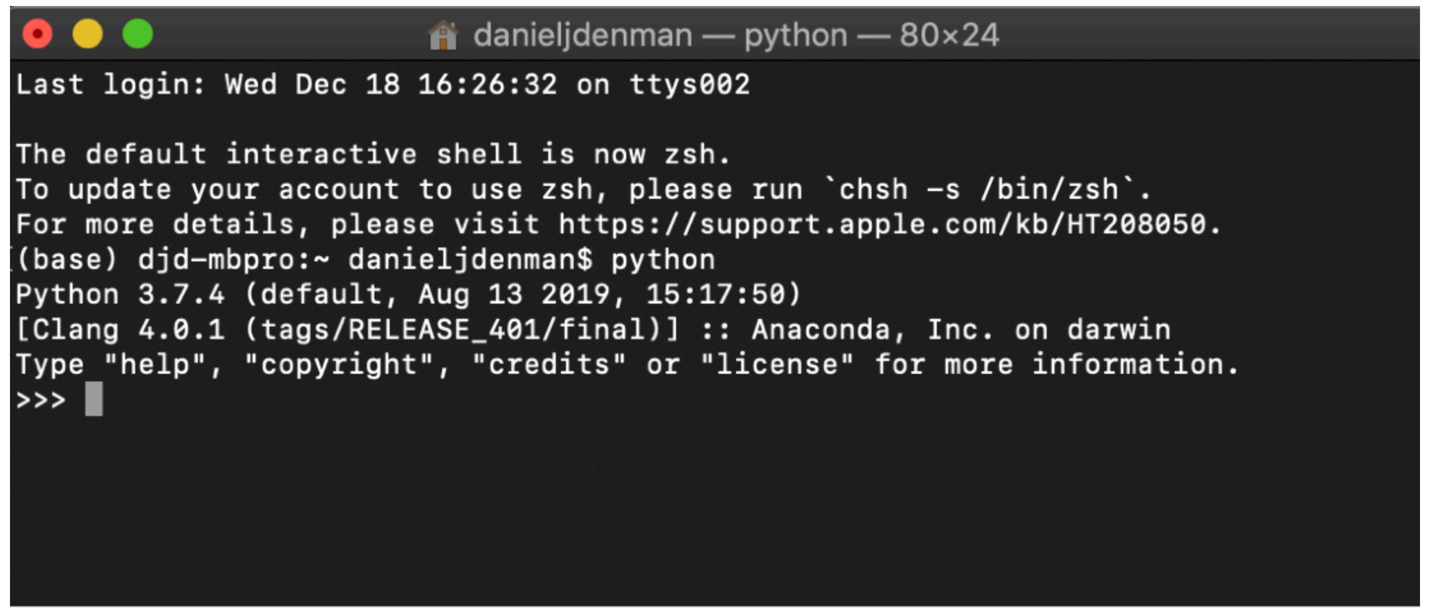

```
danieljdenman — python — 80×24
Last login: Wed Dec 18 16:26:32 on ttys002

The default interactive shell is now zsh.
To update your account to use zsh, please run `chsh -s /bin/zsh`.
For more details, please visit https://support.apple.com/kb/HT208050.
(base) djd-mbpro:~ danieljdenman$ python
Python 3.7.4 (default, Aug 13 2019, 15:17:50)
[Clang 4.0.1 (tags/RELEASE_401/final)] :: Anaconda, Inc. on darwin
Type "help", "copyright", "credits" or "license" for more information.
>>>
```

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Package Managers
Environments

ANACONDA®

Containerized

docker

COCALC

overview: levels

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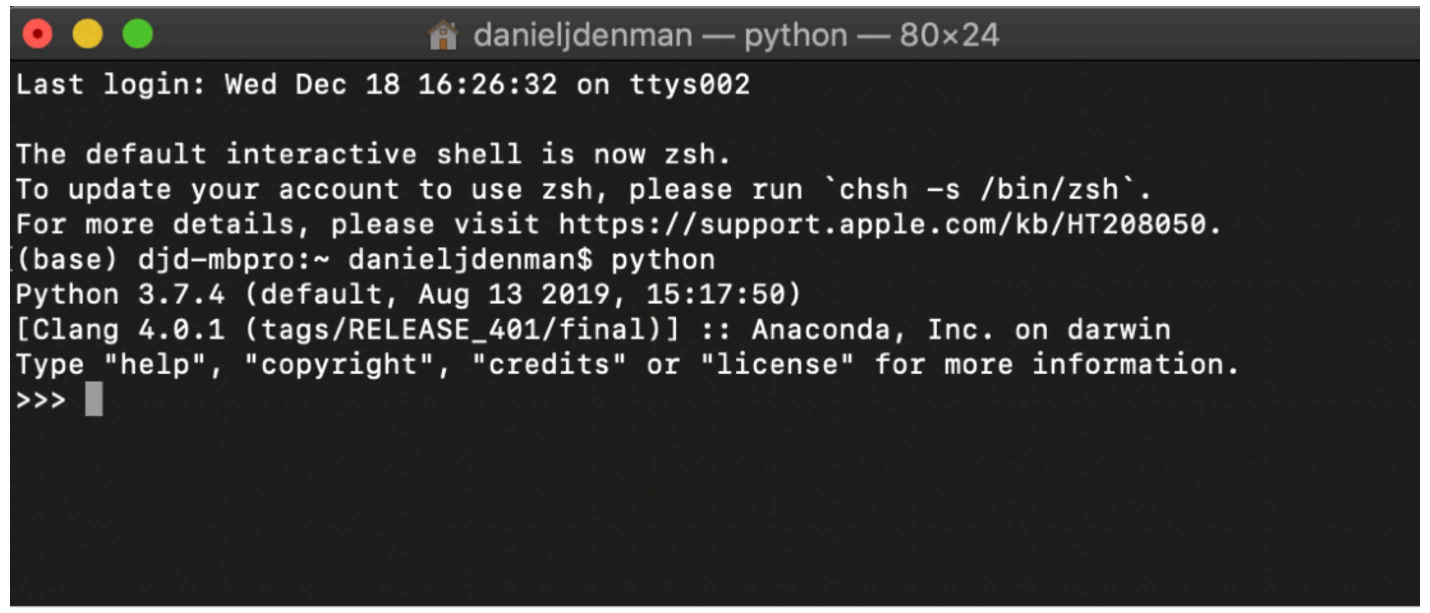

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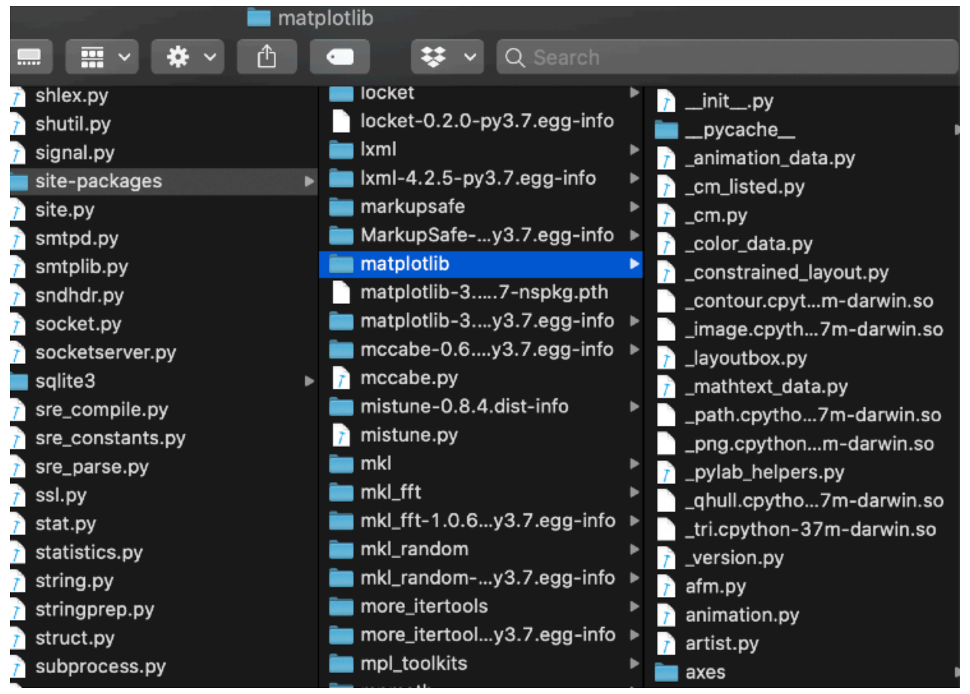

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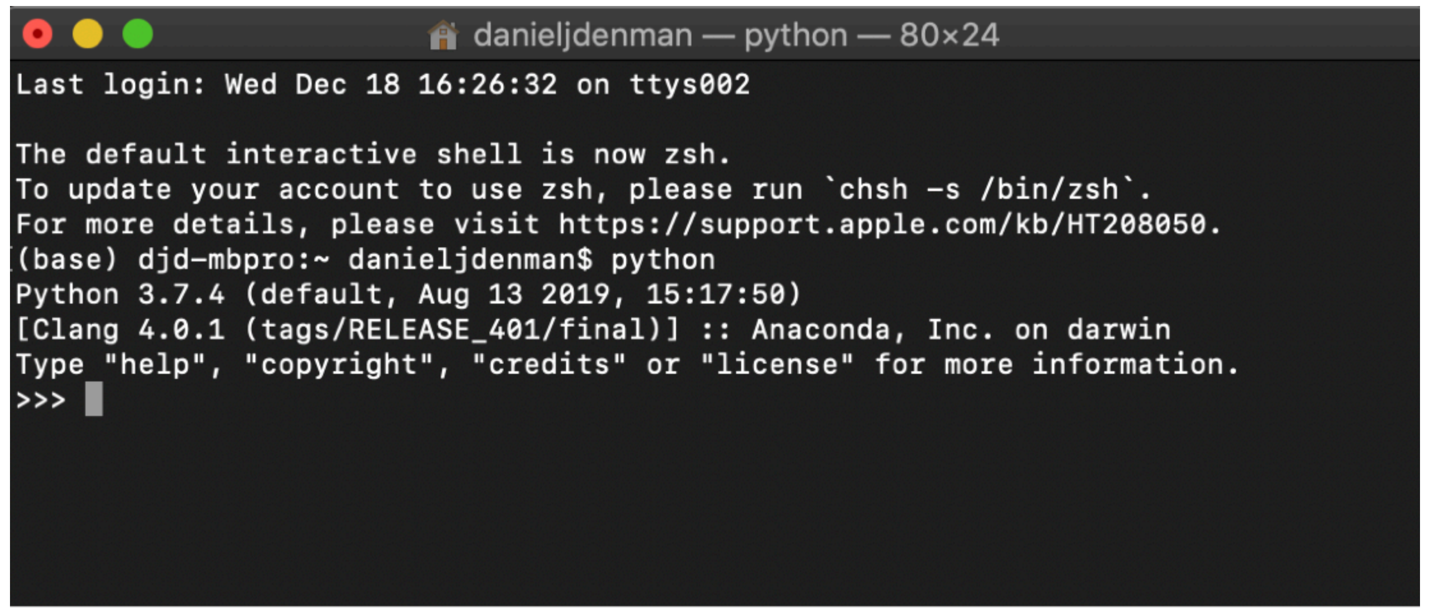

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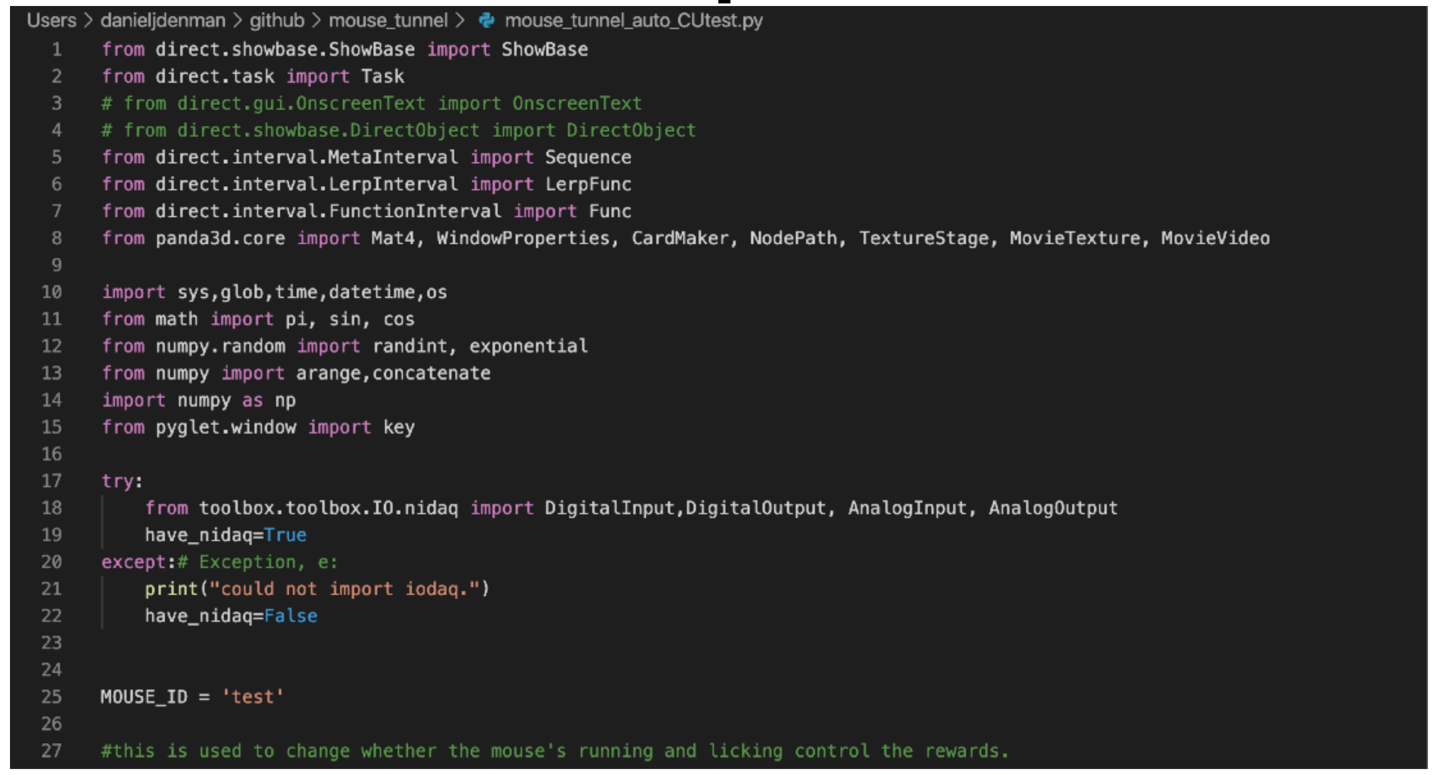

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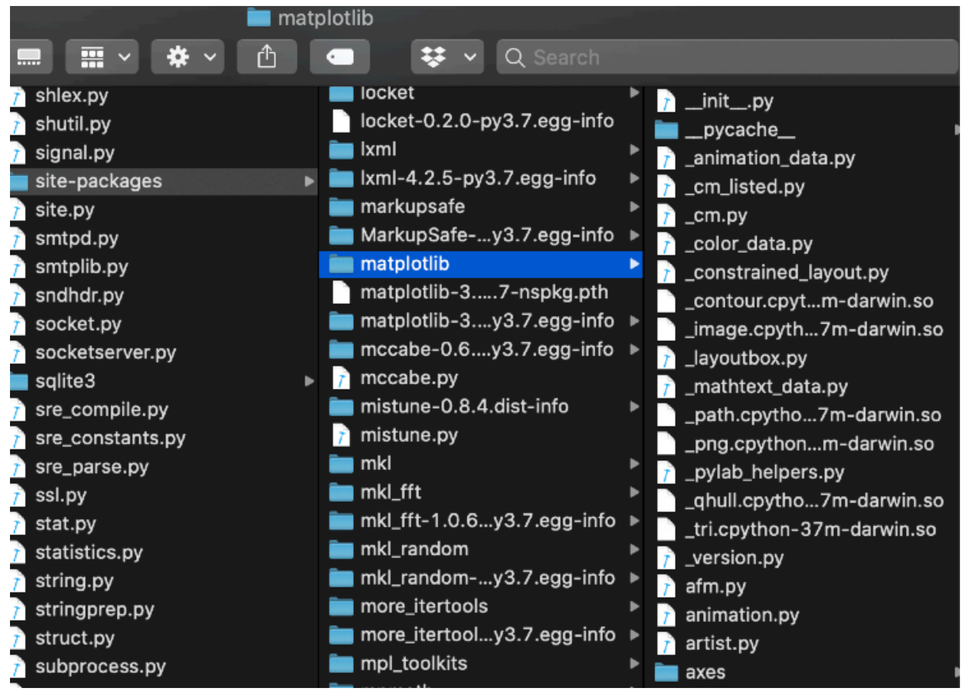

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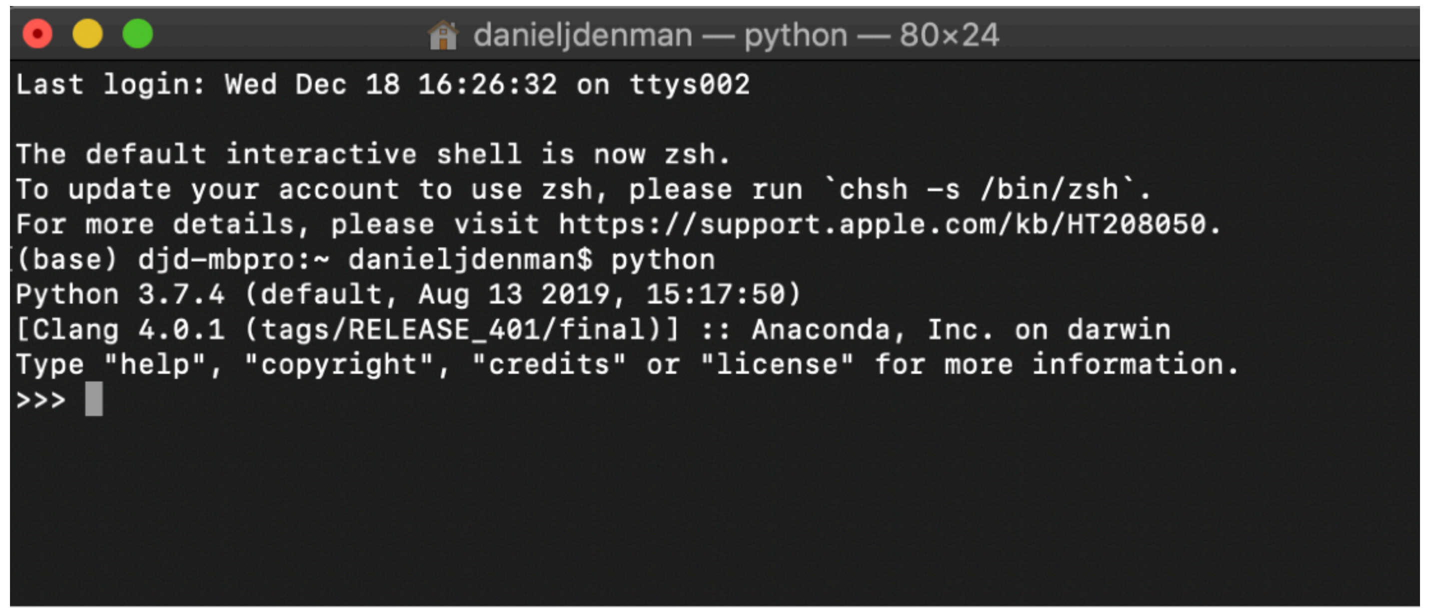

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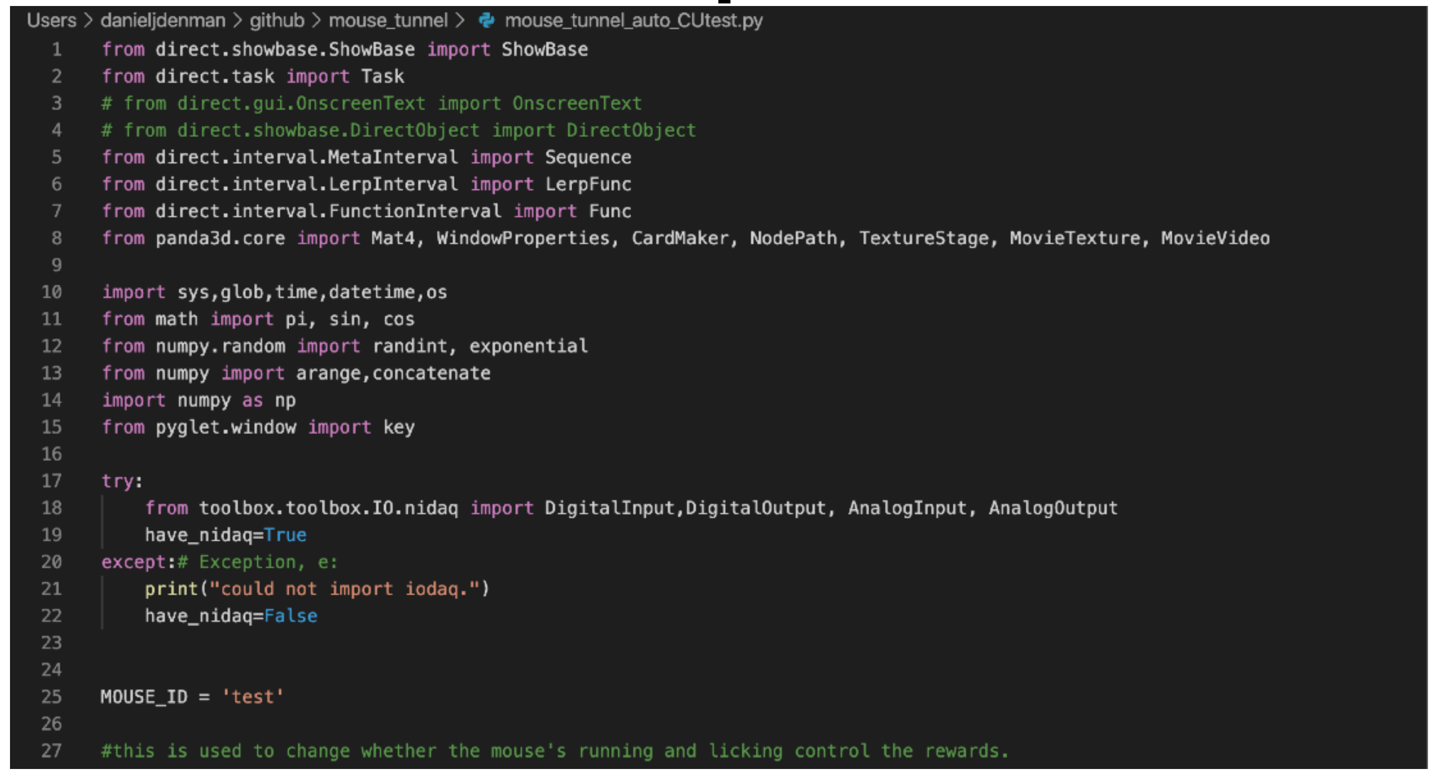

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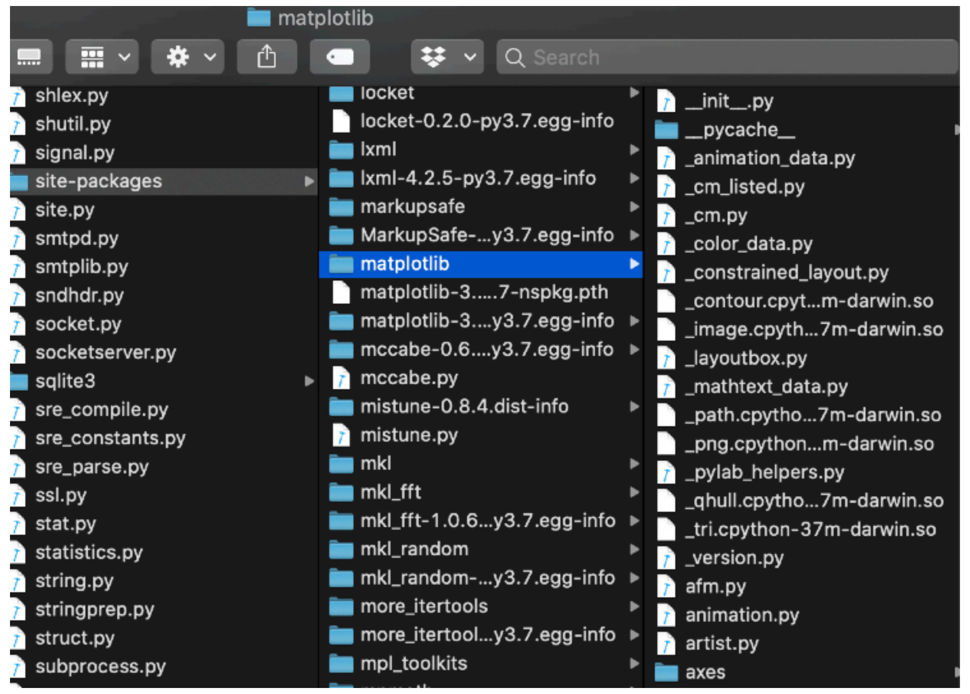

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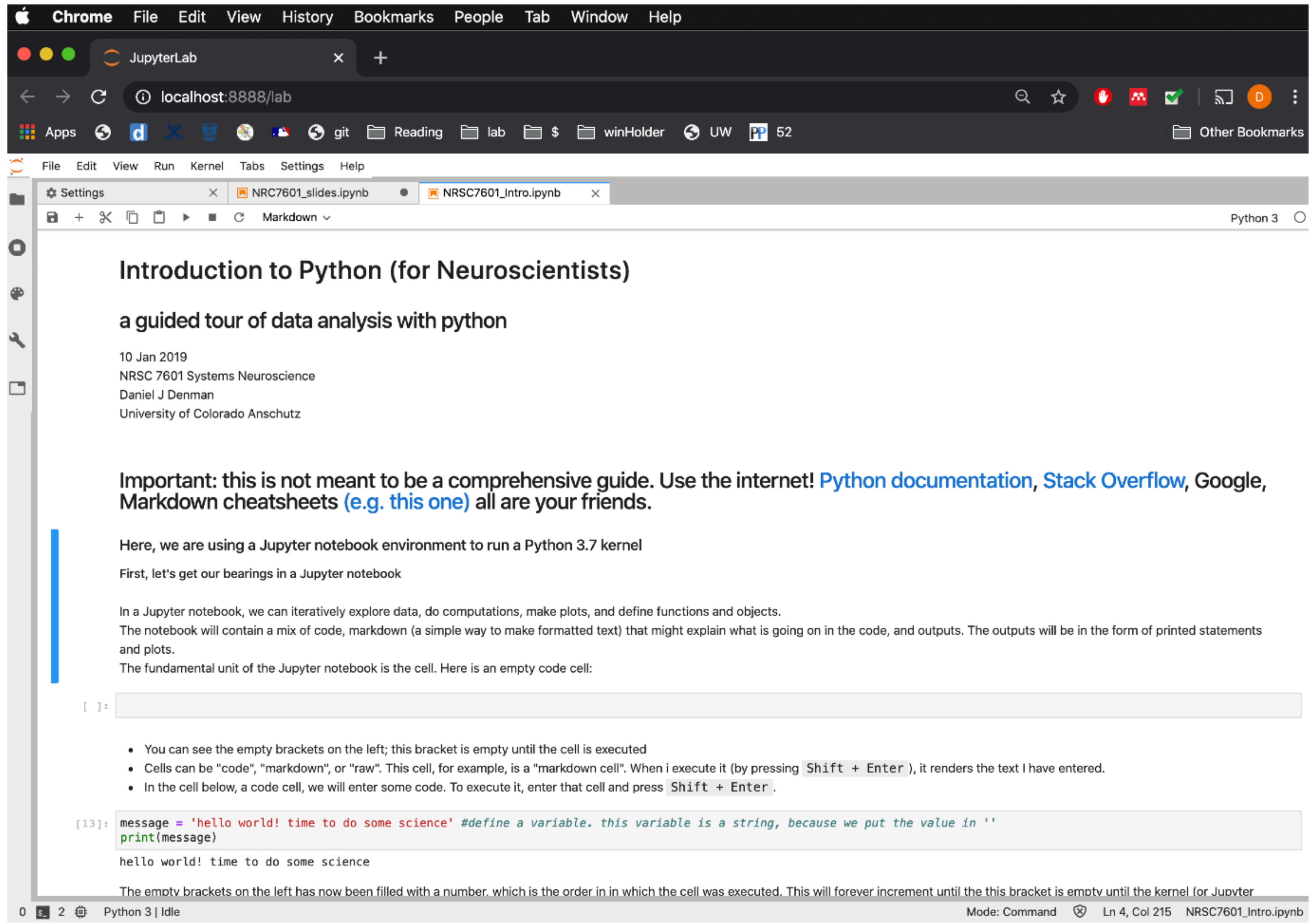

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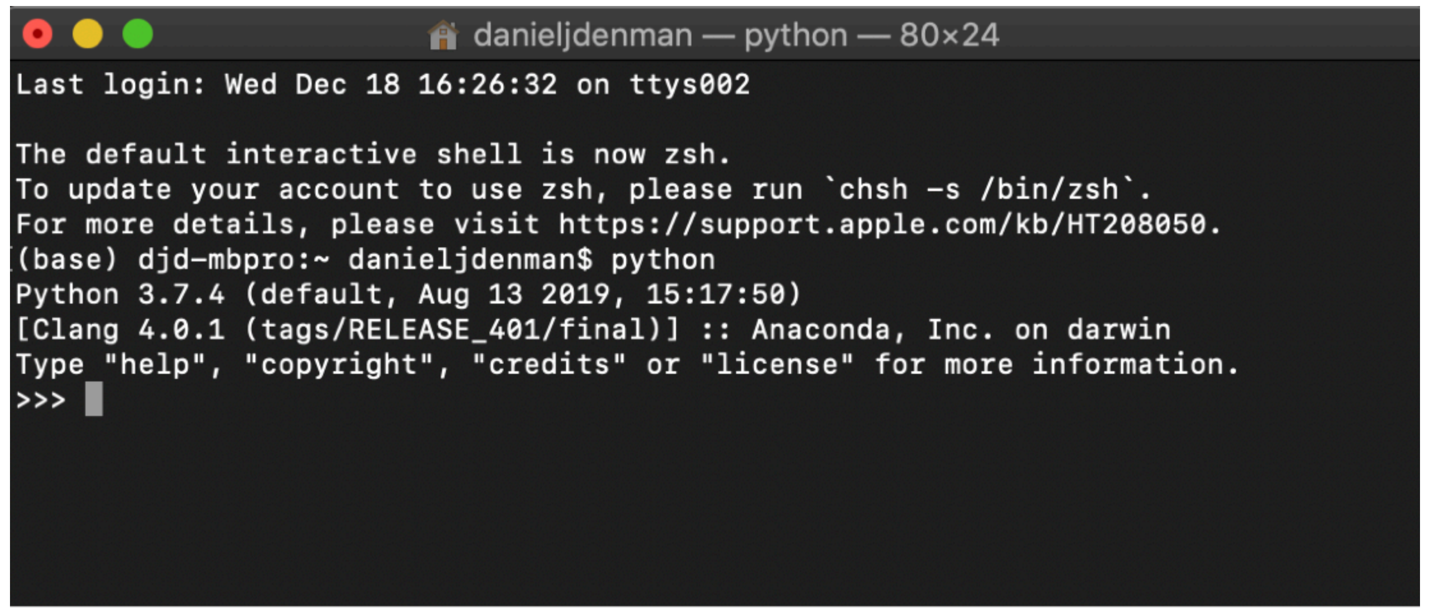

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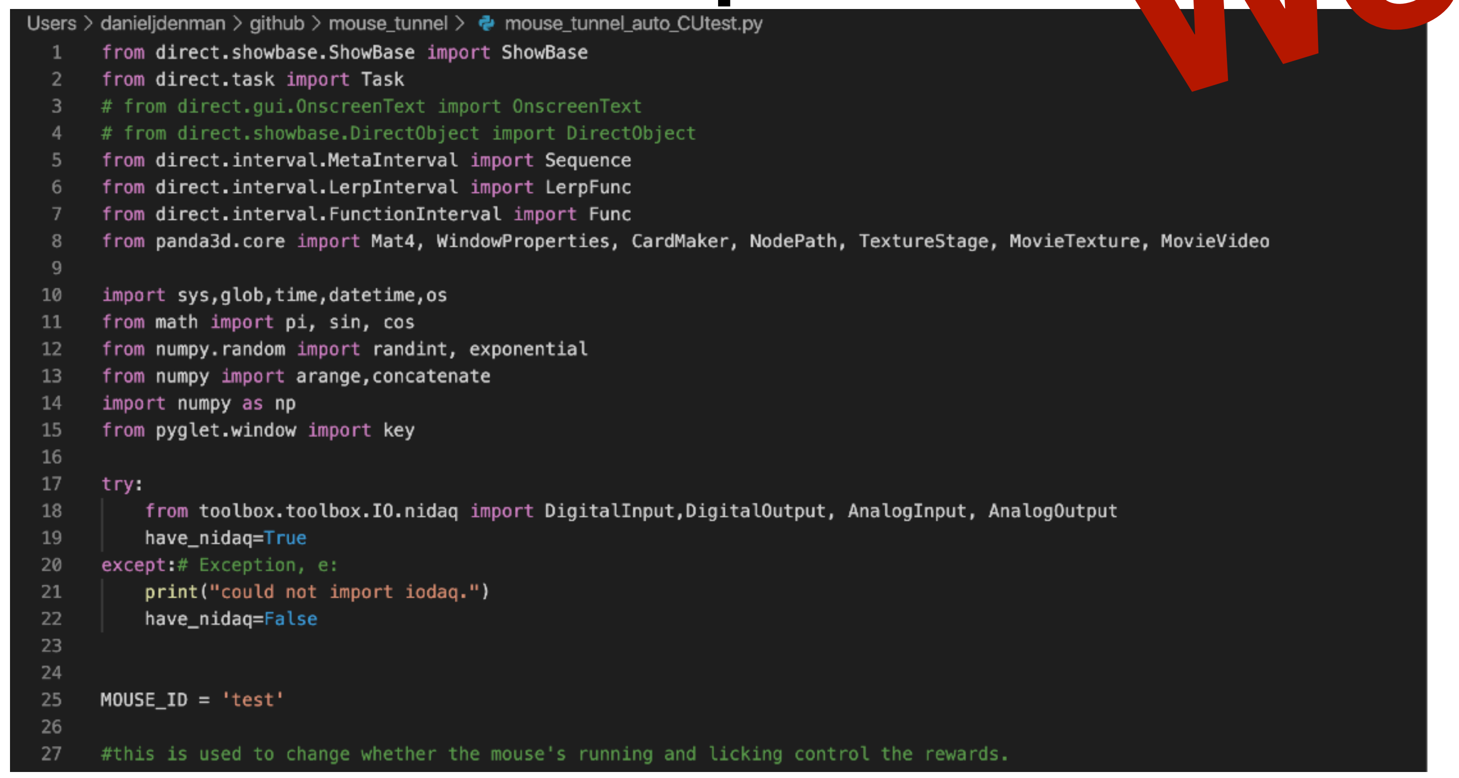

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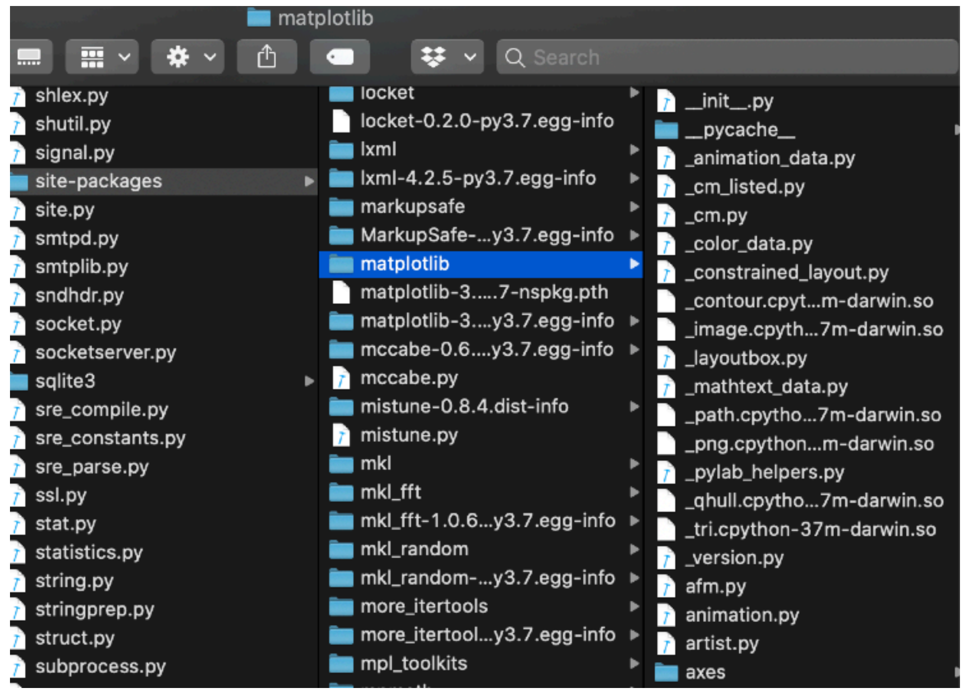

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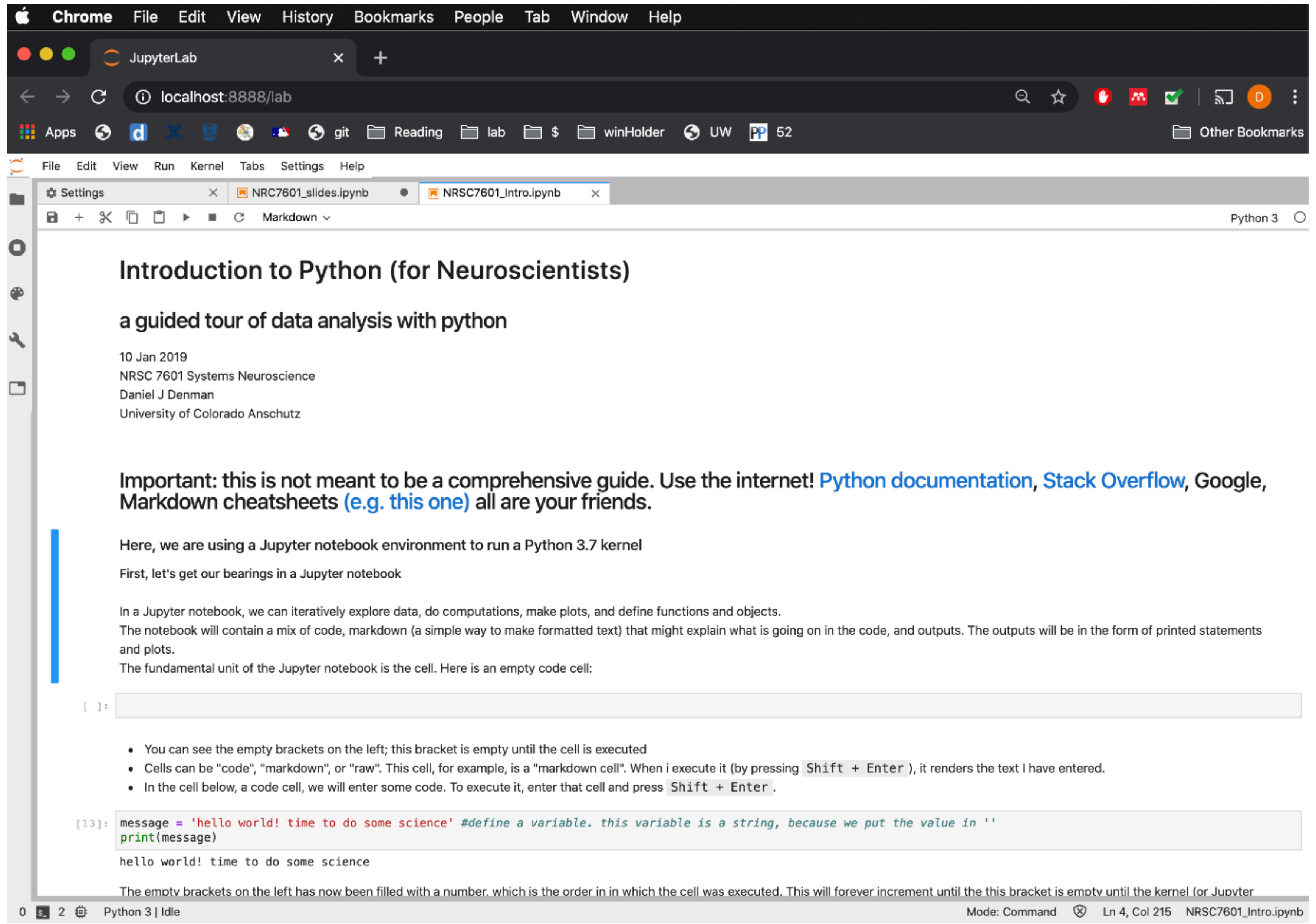

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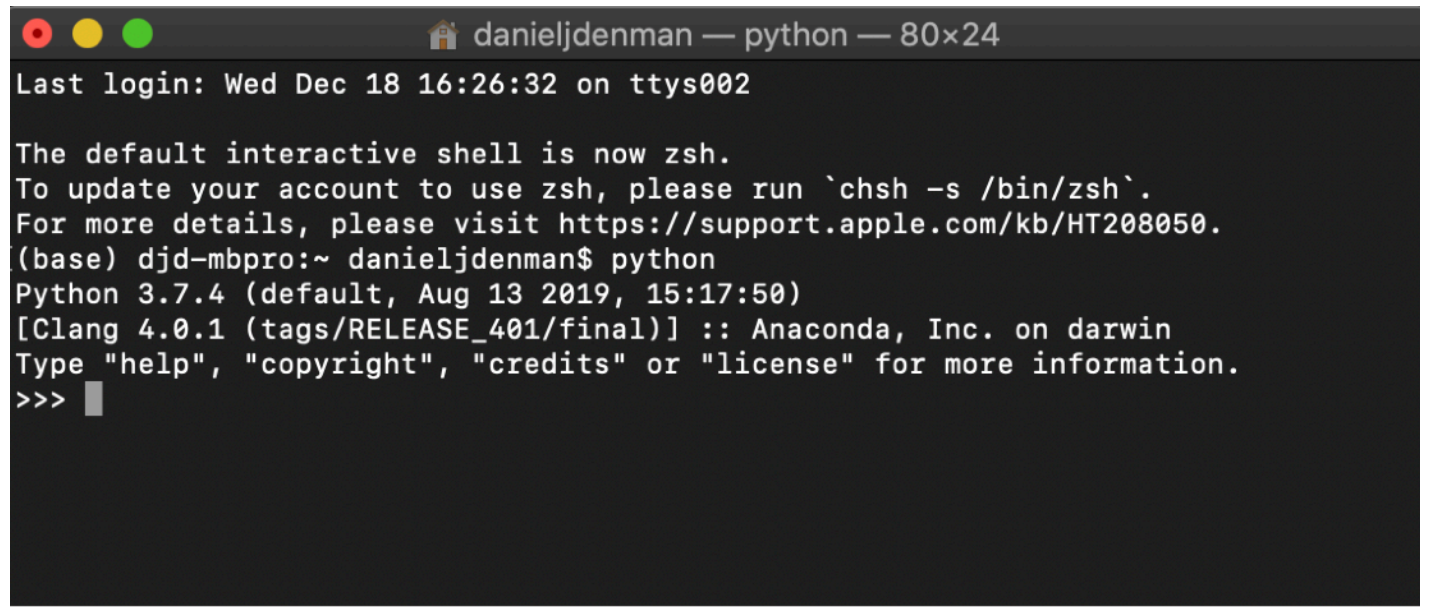

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Week 6

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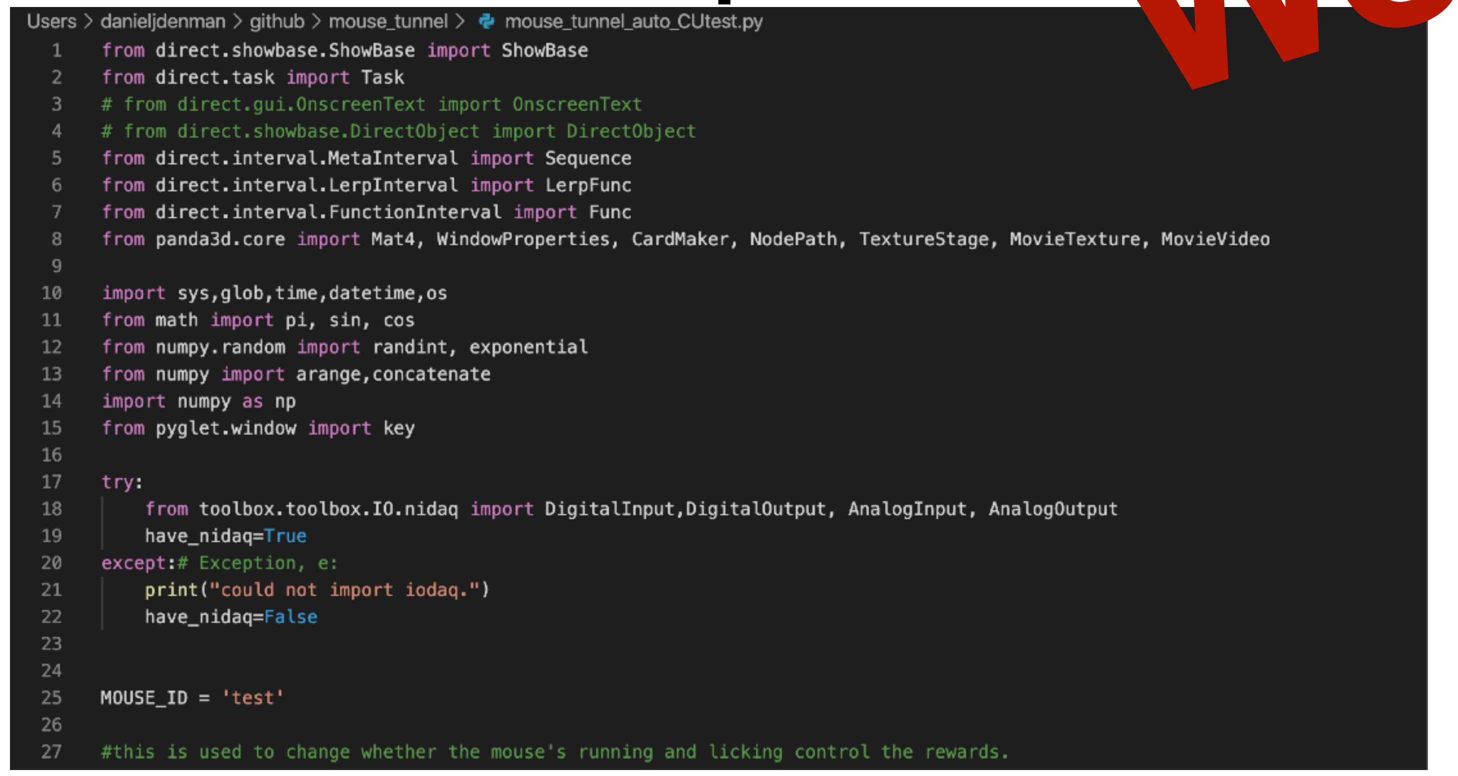

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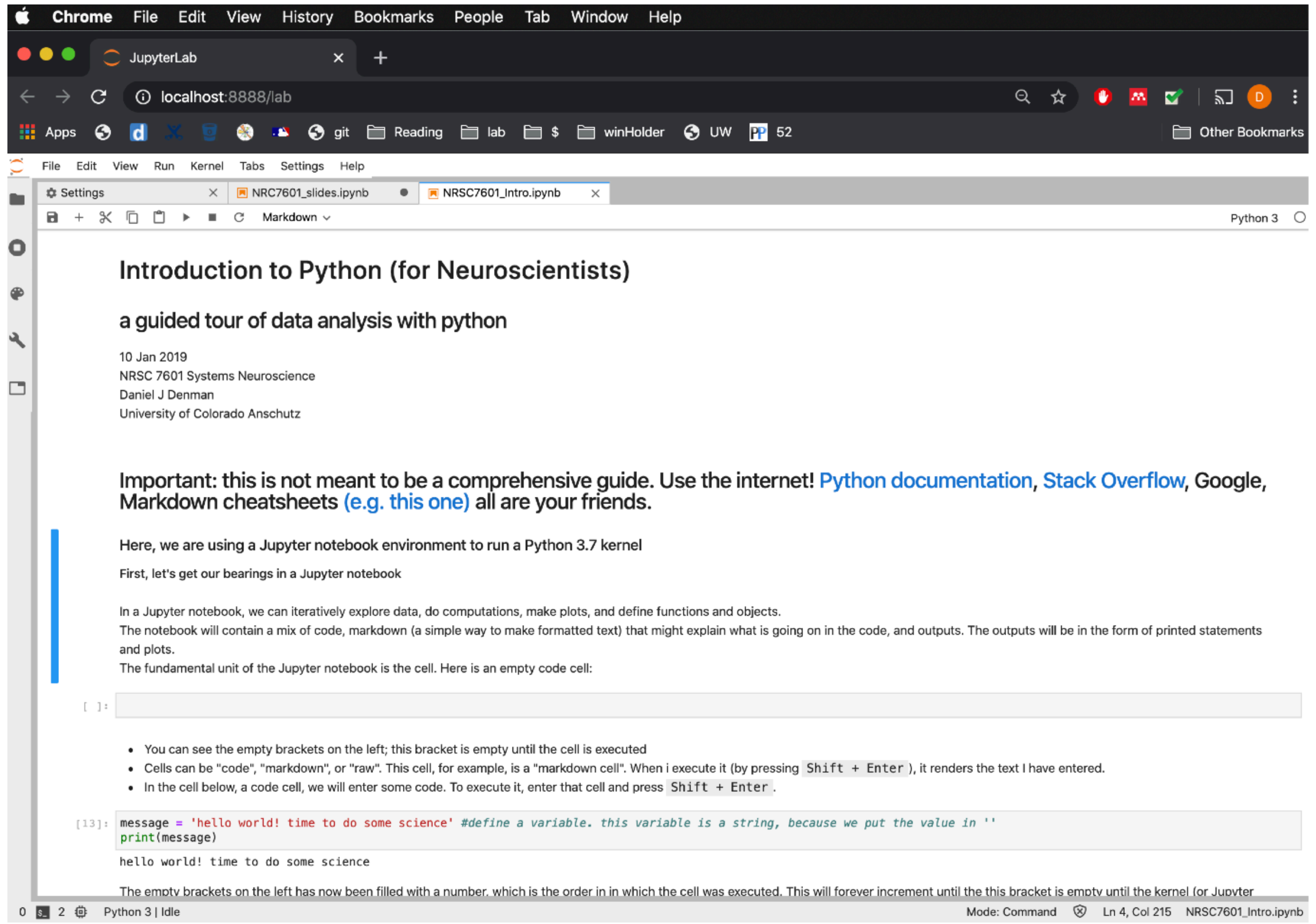

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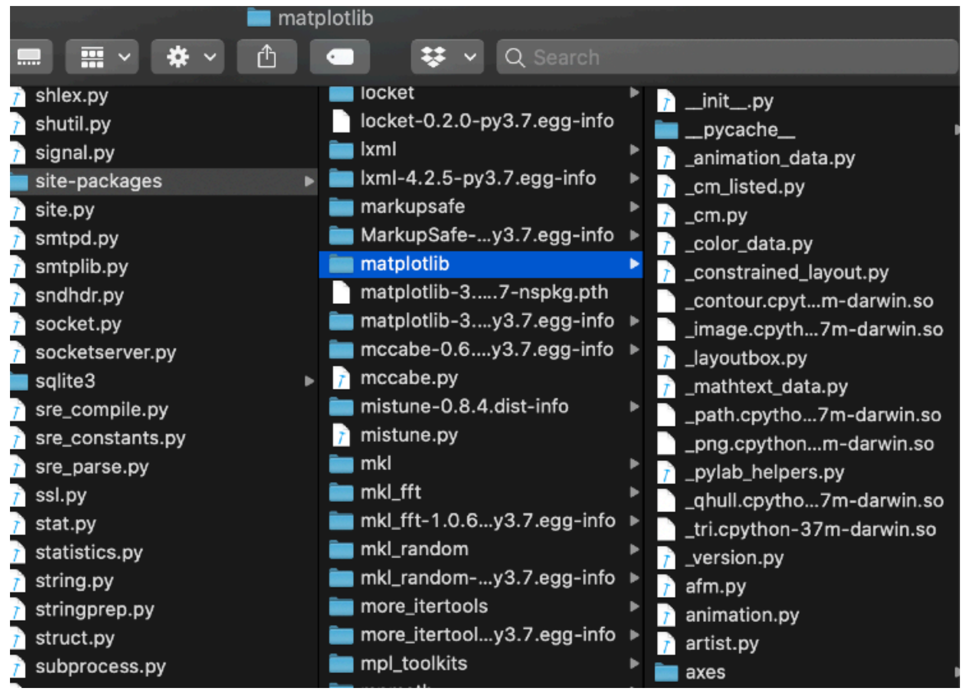

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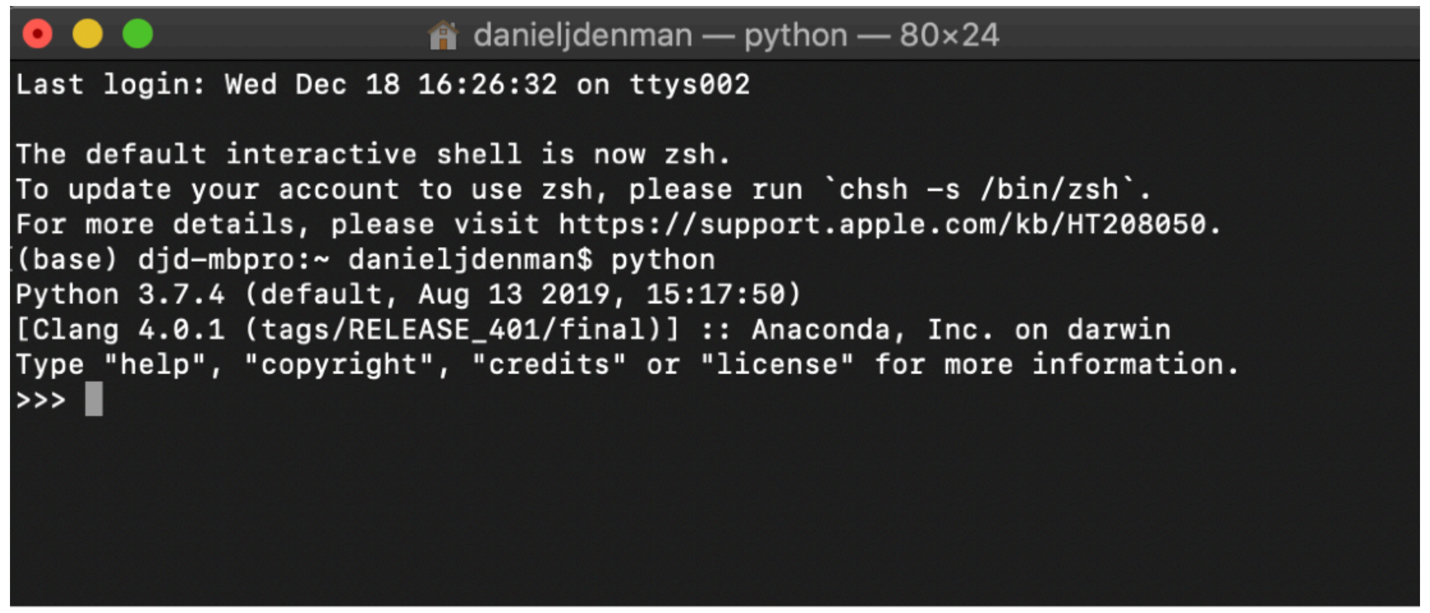

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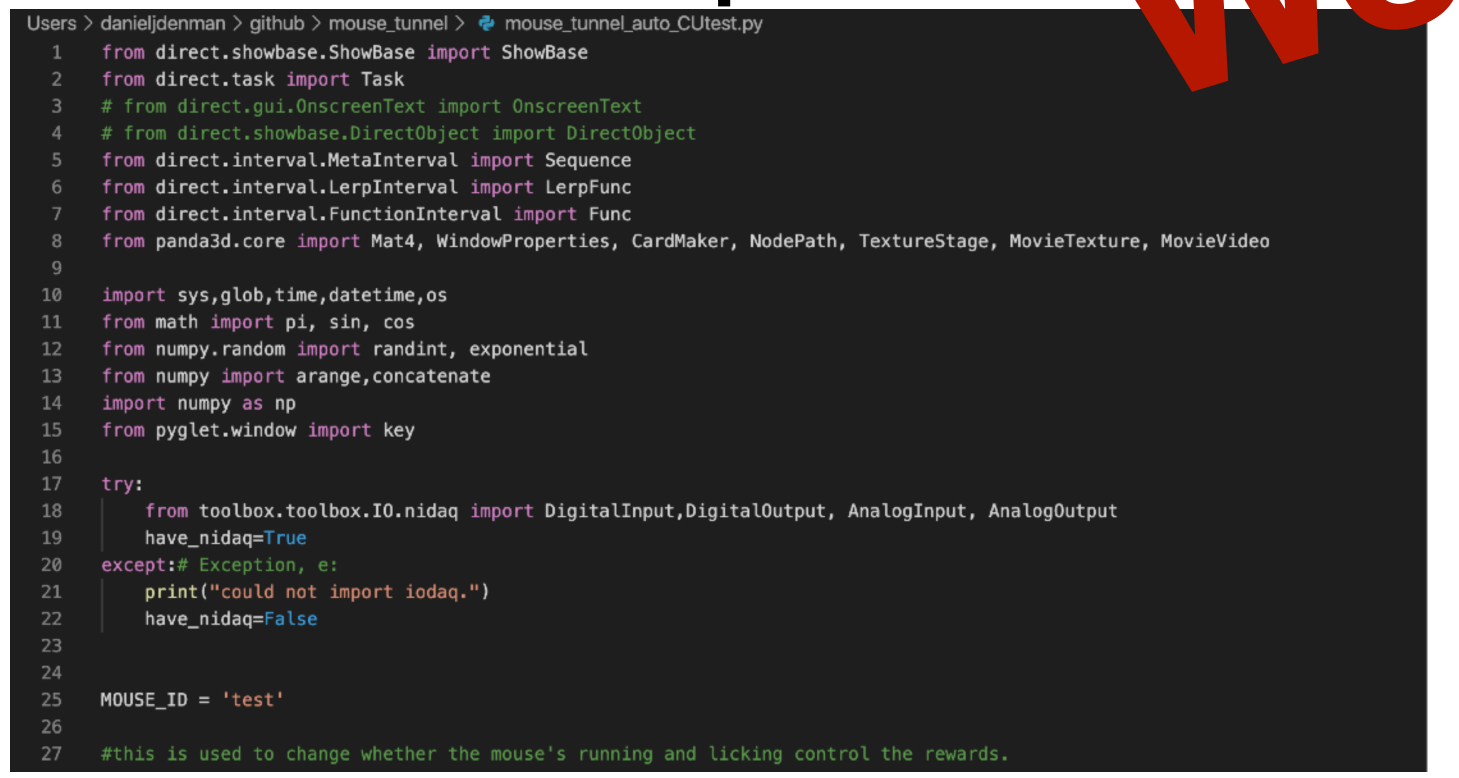

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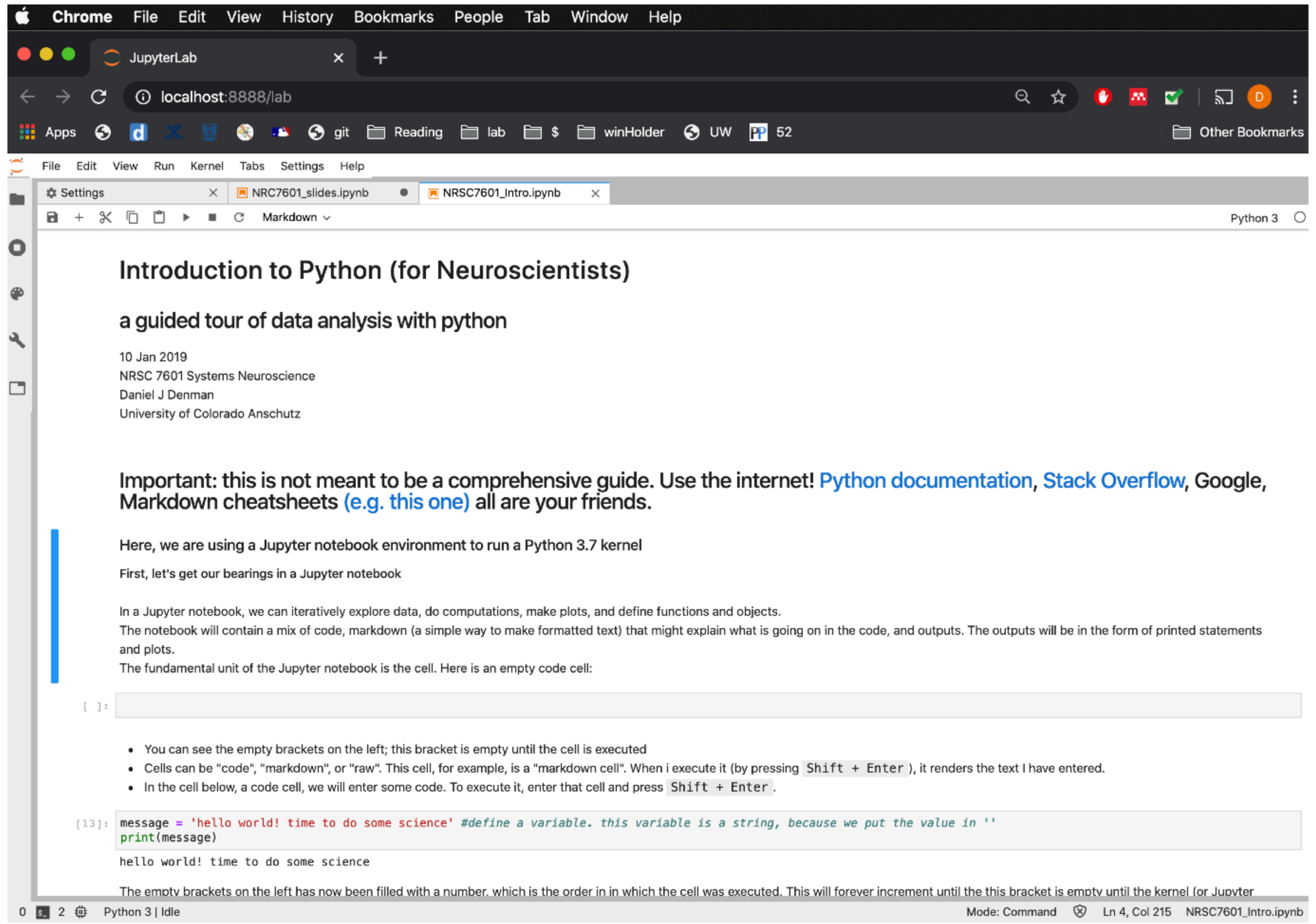

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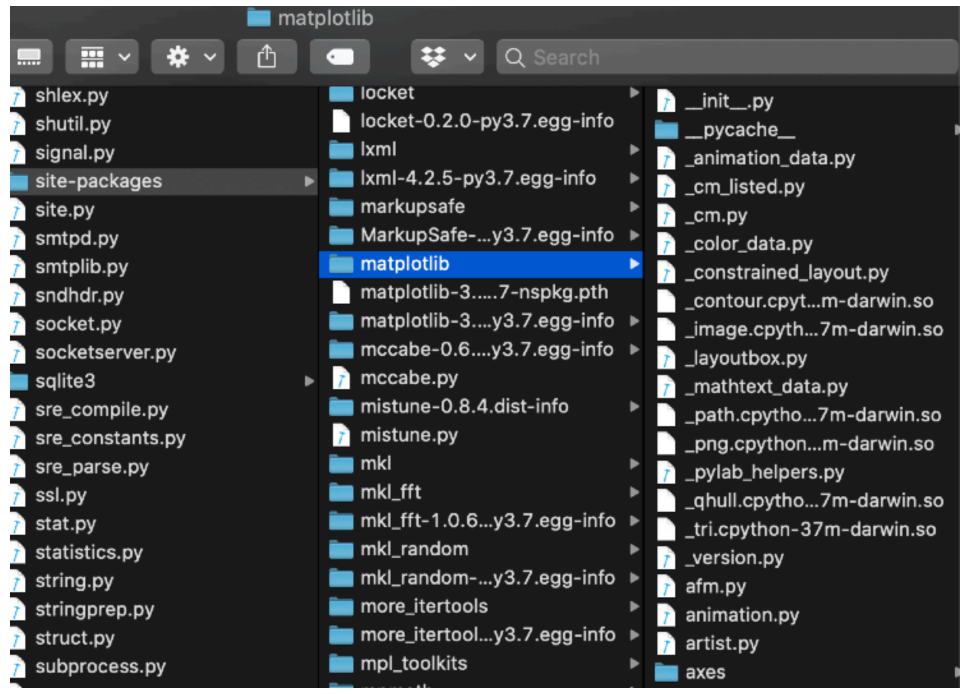

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Stack Overflow <— not cheating!

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Hardware control

RasberryPi
Arduino
PyDAQMX
PsychoPy
…many APIs…

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<u>**packages!**</u>

Stack Overflow <— not cheating!

also it is free —> democratizing science

in this realm, cloud resources (data, compute) also open science to a wider group that aren't collecting their own data and running their own super computers

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Week 5
Week 6

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Weeks 7-9

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Week 5
Week 6

MATLAB

- At this time, some understanding of both Python and MATLAB is extremely useful. We use both in the Denman Lab, but my primary expertise is in Python
- John T going to be MATLAB point person for the course

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Week 1

- **Anaconda environments and where python is on your computer**
- **VS Code, starting a jupyter notebook, running a script**
- **Basics and syntax review**

Style and philosophy

- **Don't:** overoptimize; get it done first
- **Do:** document as you go
- Pay attention to variable names
- There is a slippery slope from the minimal functional code to unusable/unshareable code
- Read this PLoS paper this week!

PLOS COMPUTATIONAL BIOLOGY

OPEN ACCESS

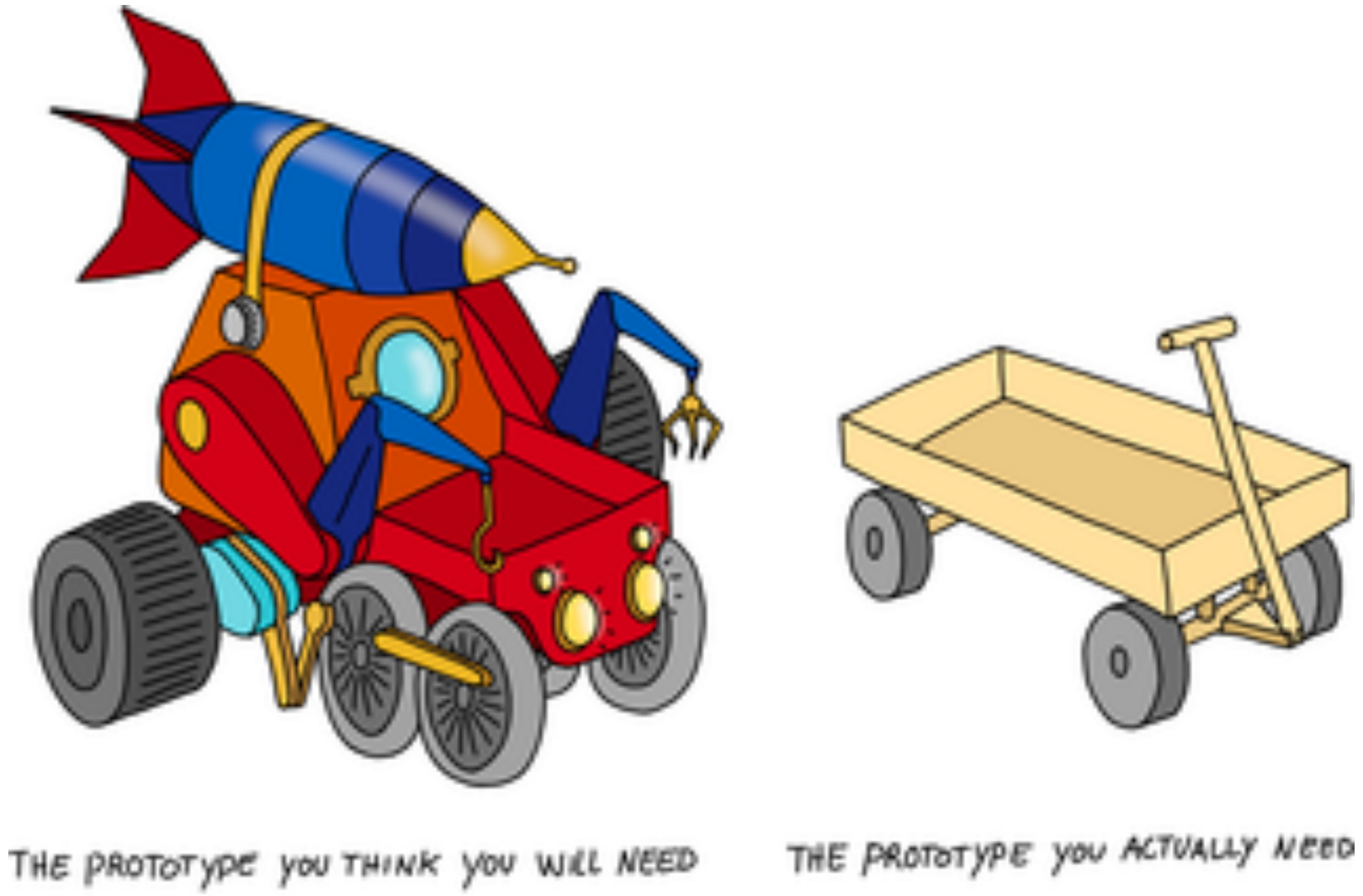
EDITORIAL

Ten simple rules for quick and dirty scientific programming

Gabriel Balaban, Ivar Grytten, Knut Dagestad Rand, Lonneke Scheffer, Geir Kjetil Sandve

Published: March 11, 2021 • https://doi.org/10.1371/journal.pcbi.1008549

https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1008549



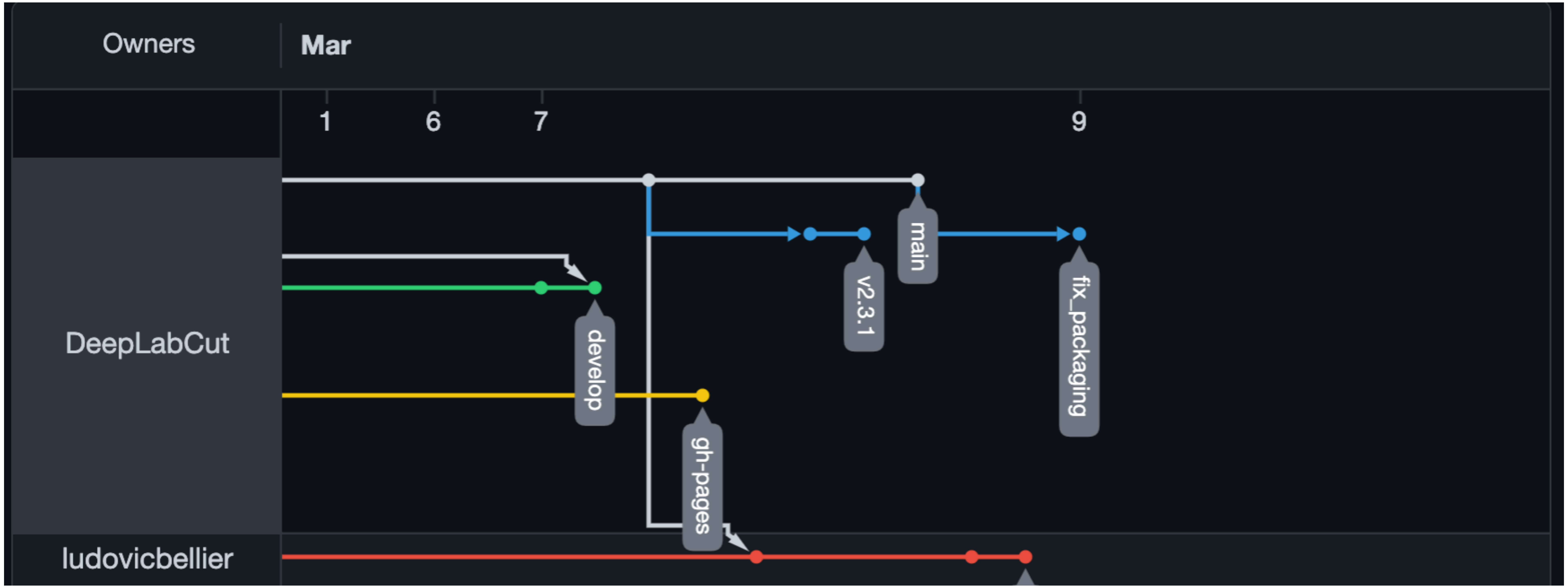
git

...and GitHub, are **version control**

- This is important. And not intuitive. It will likely make you frustrated and/or confused at some point.
- Version control is not optional; if you don't use git for version control, you are going to use something else (e.g,: analysis_script_v1.py, analysis_script_v2.py, analysis_script_v2_20210622.py, analysis_script_v3_07142021.py, analysis_script_final.py,analysis_script_final2.py,…, analysis_script_final2_for.py)
- Making git a part of your workflow can simplify and provide redundancy and flexibility; more advanced features also makes sharing simpler. Evaluation.
- git has to be installed, which we can use Anaconda to do (even if you are using MATLAB only for your project)
- Command-line is great, but if you are new to git we recommend using VS Code or GitHub Desktop
- **We're going to go over some git interactively to get course materials today.**

git

Conceptually

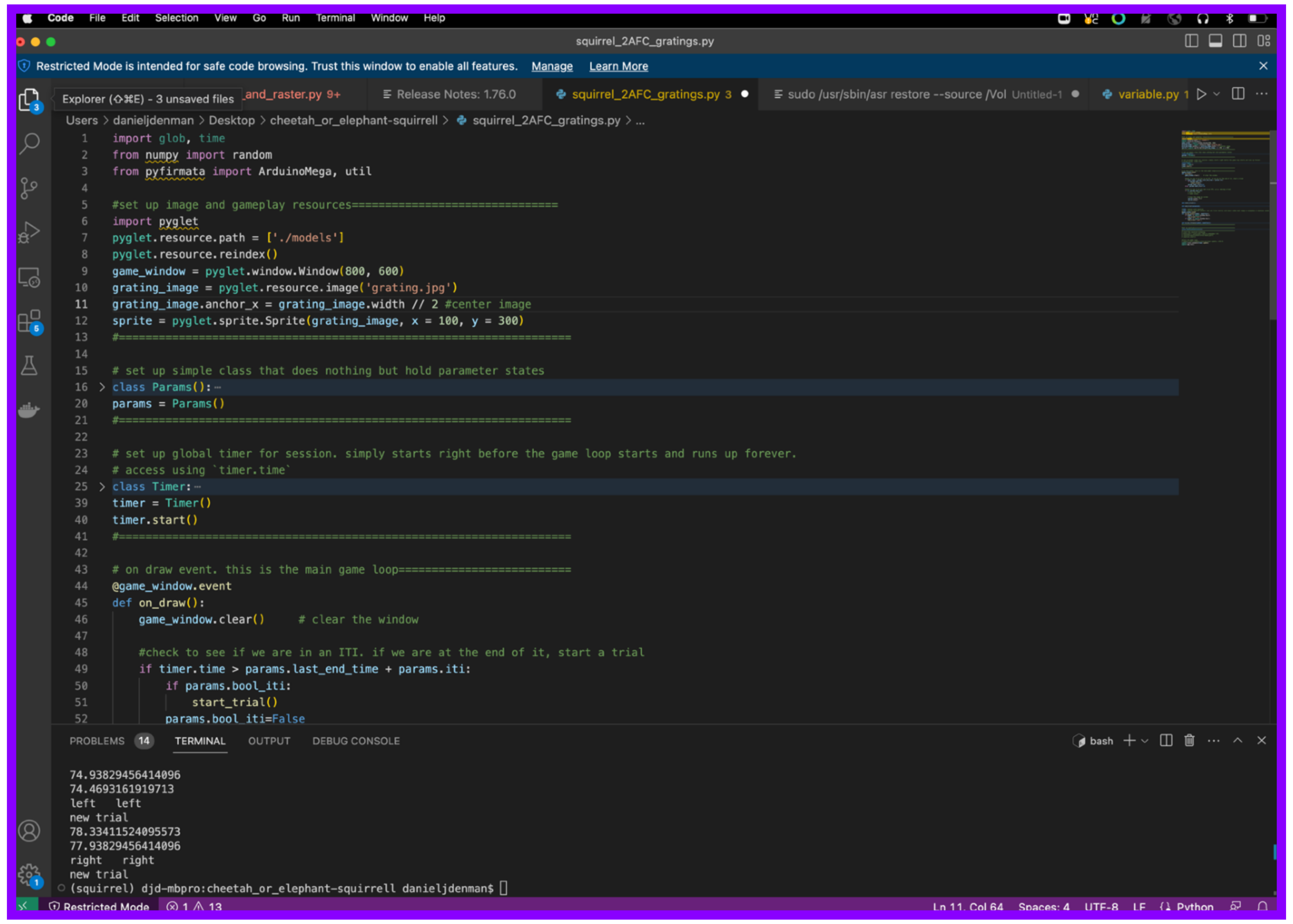


GitHub

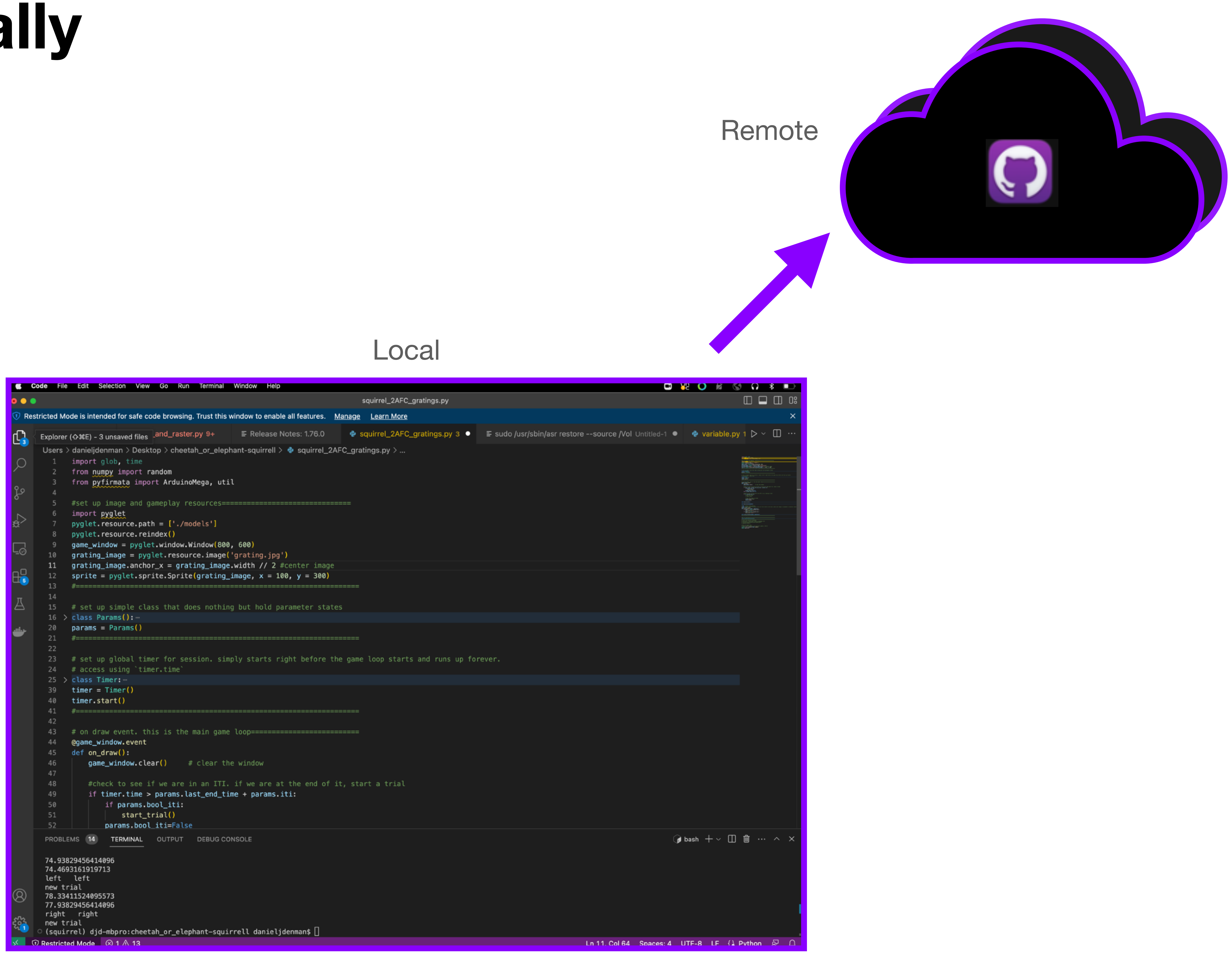
Conceptually

Remote

Local



GitHub
Conceptually



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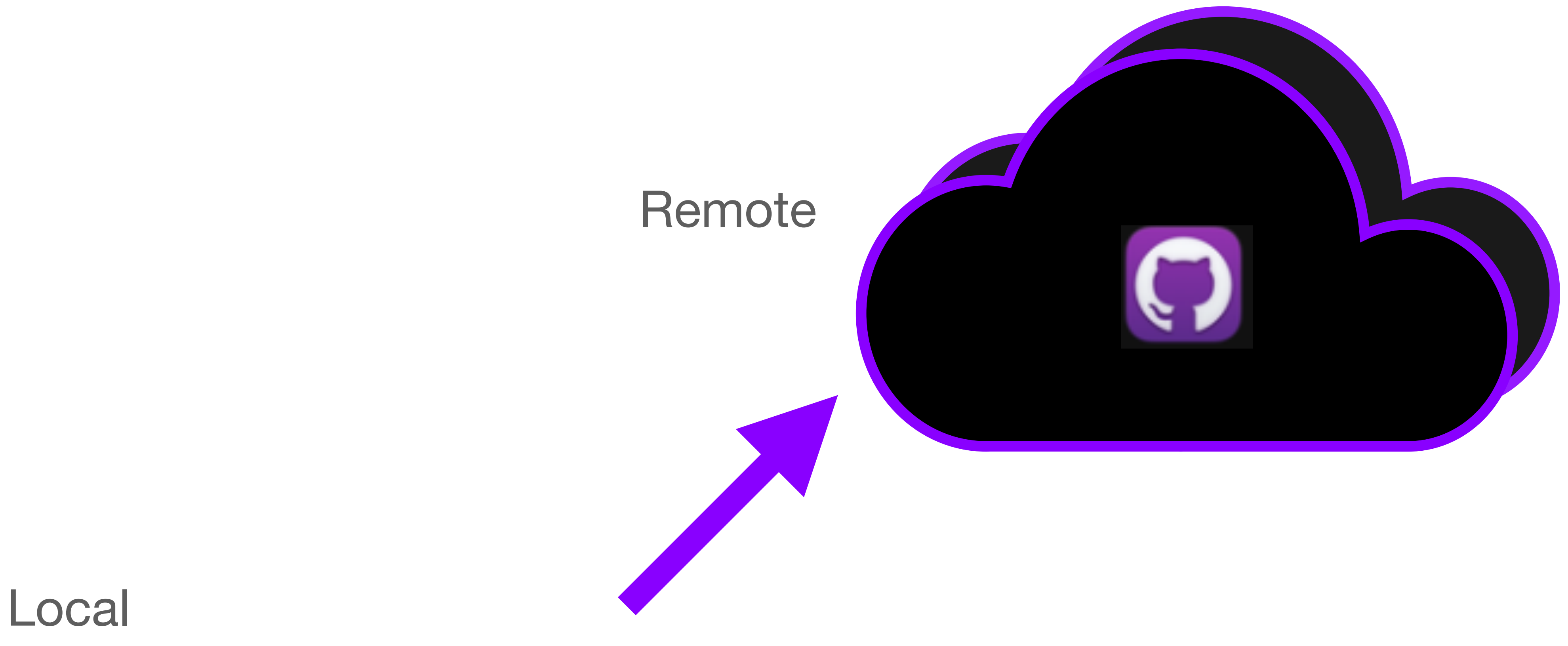

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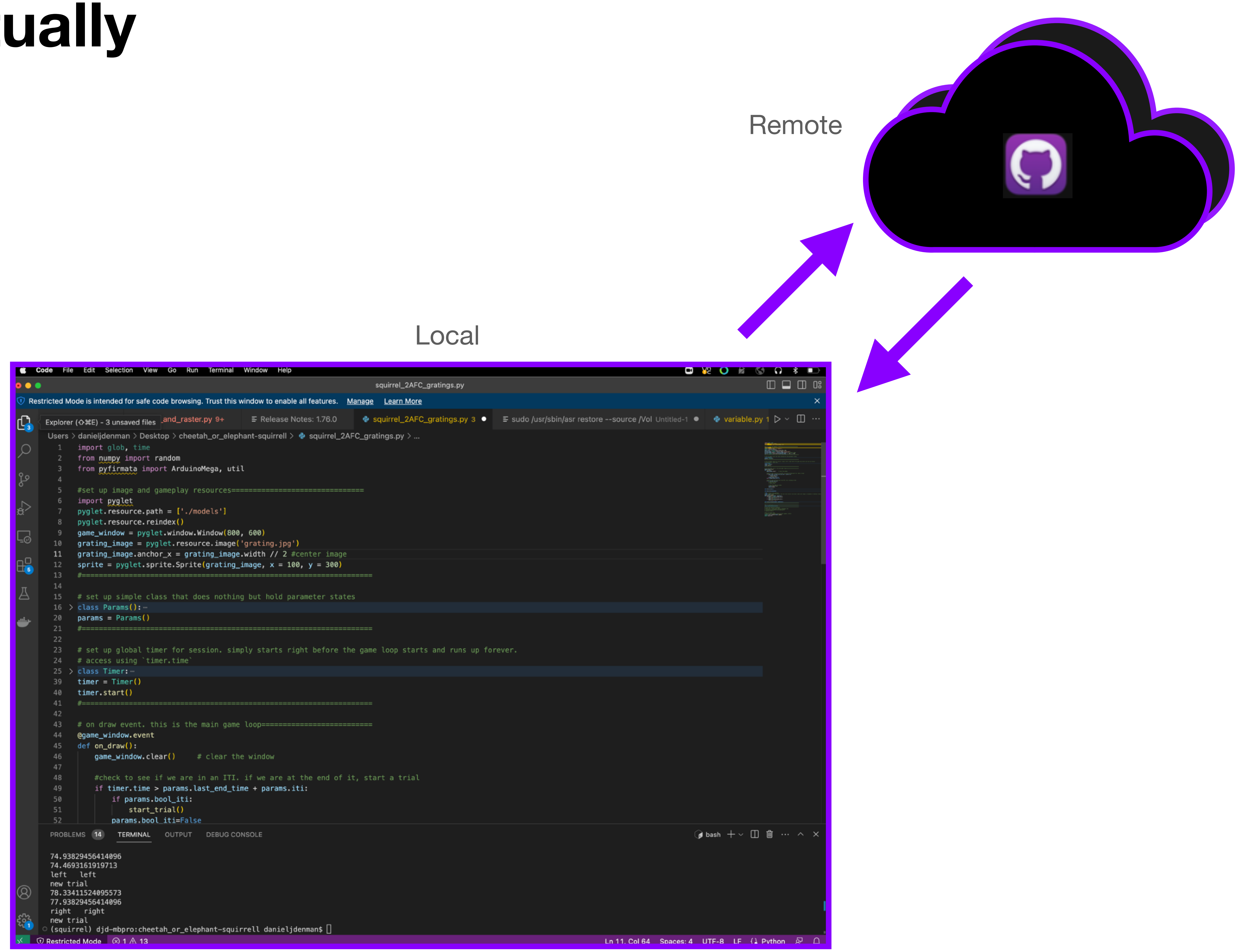

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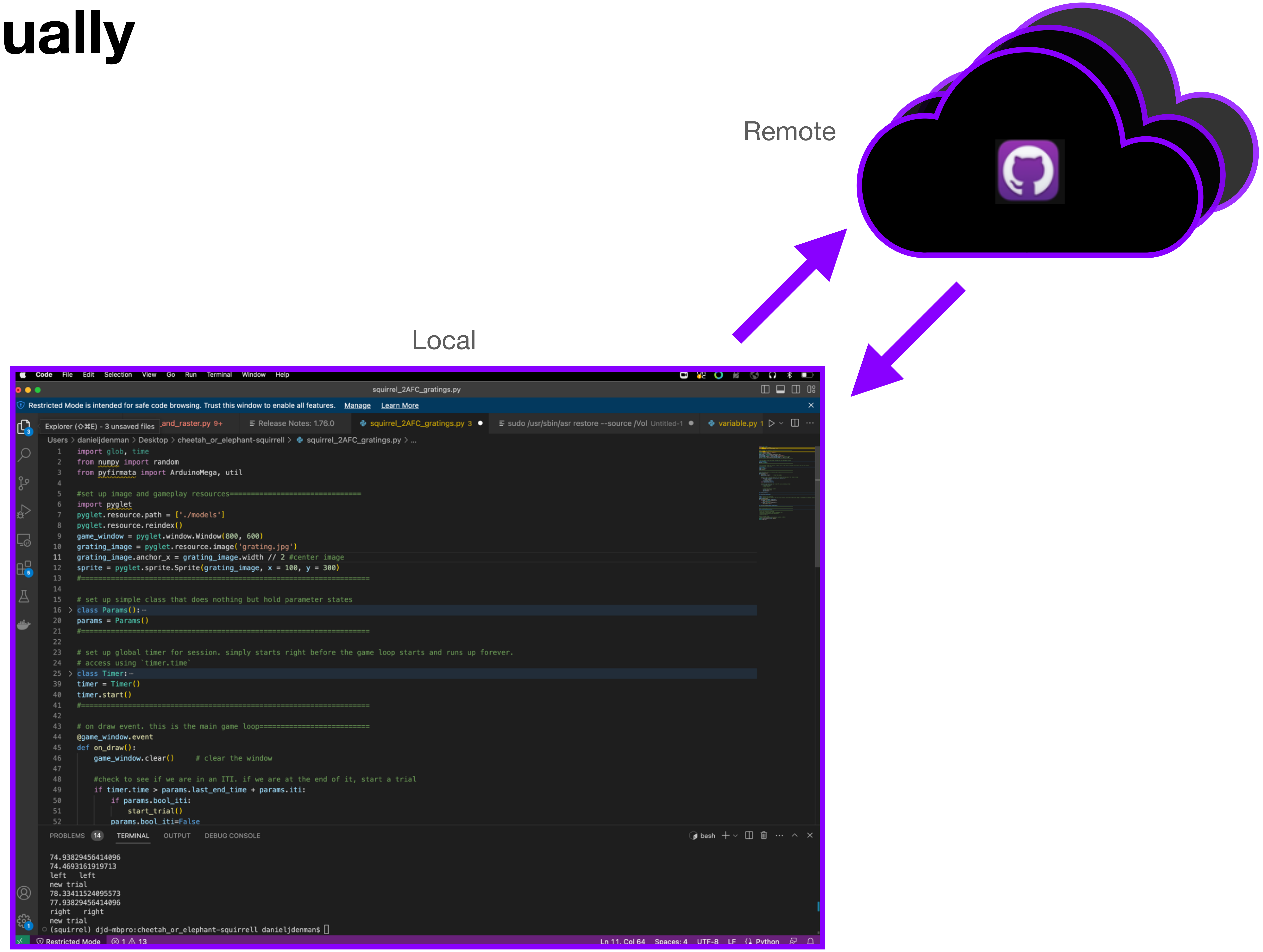

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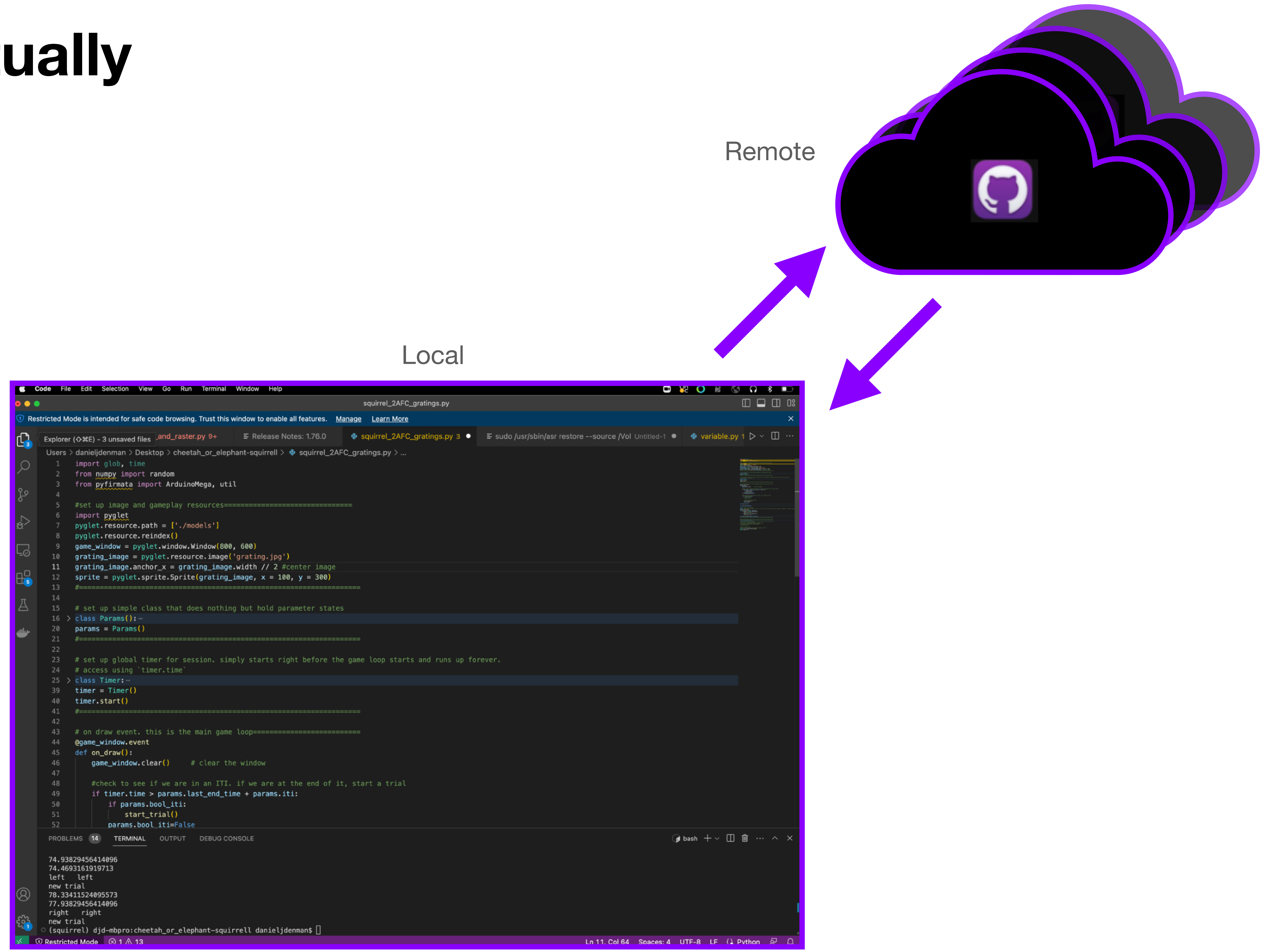

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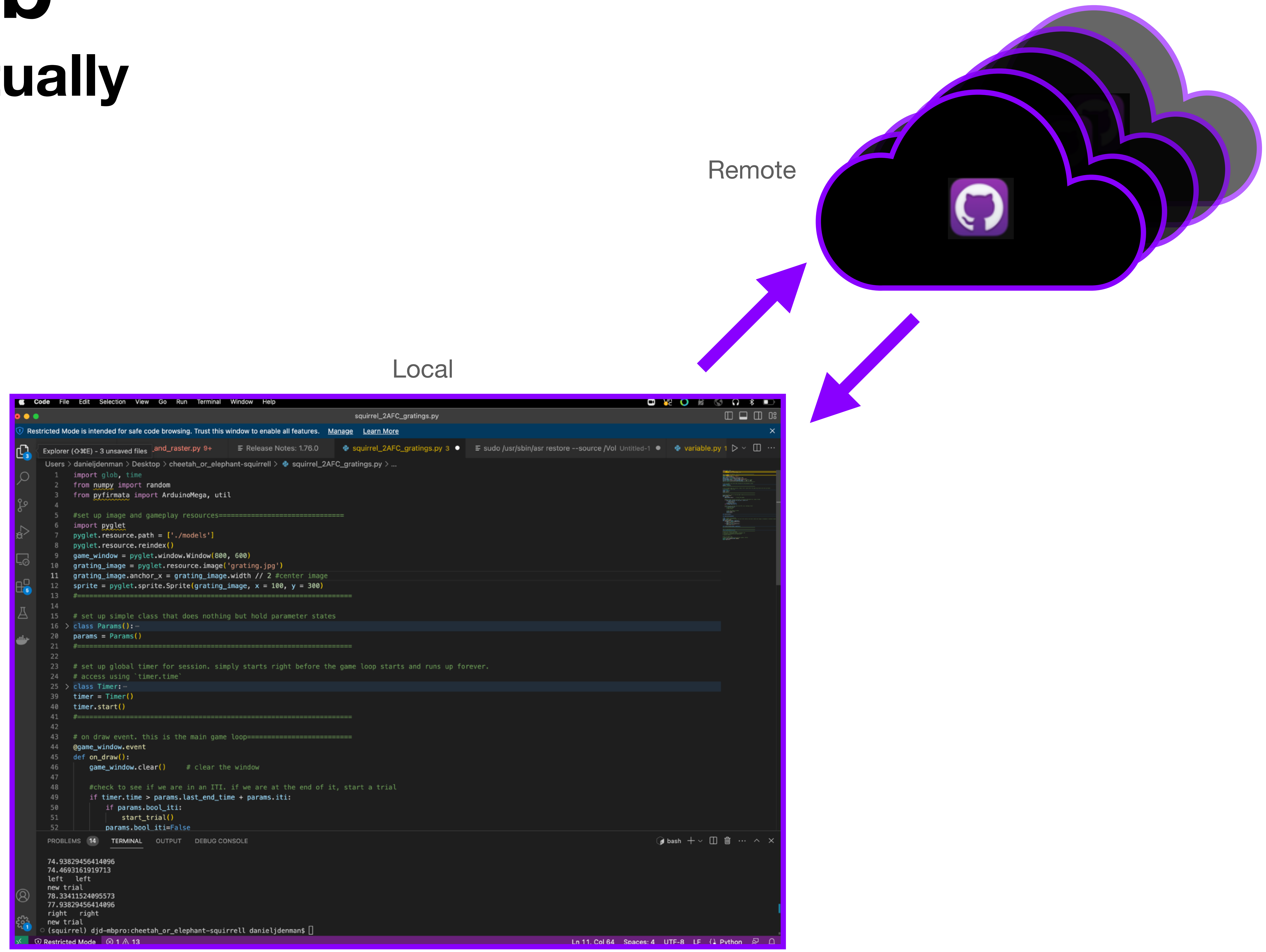

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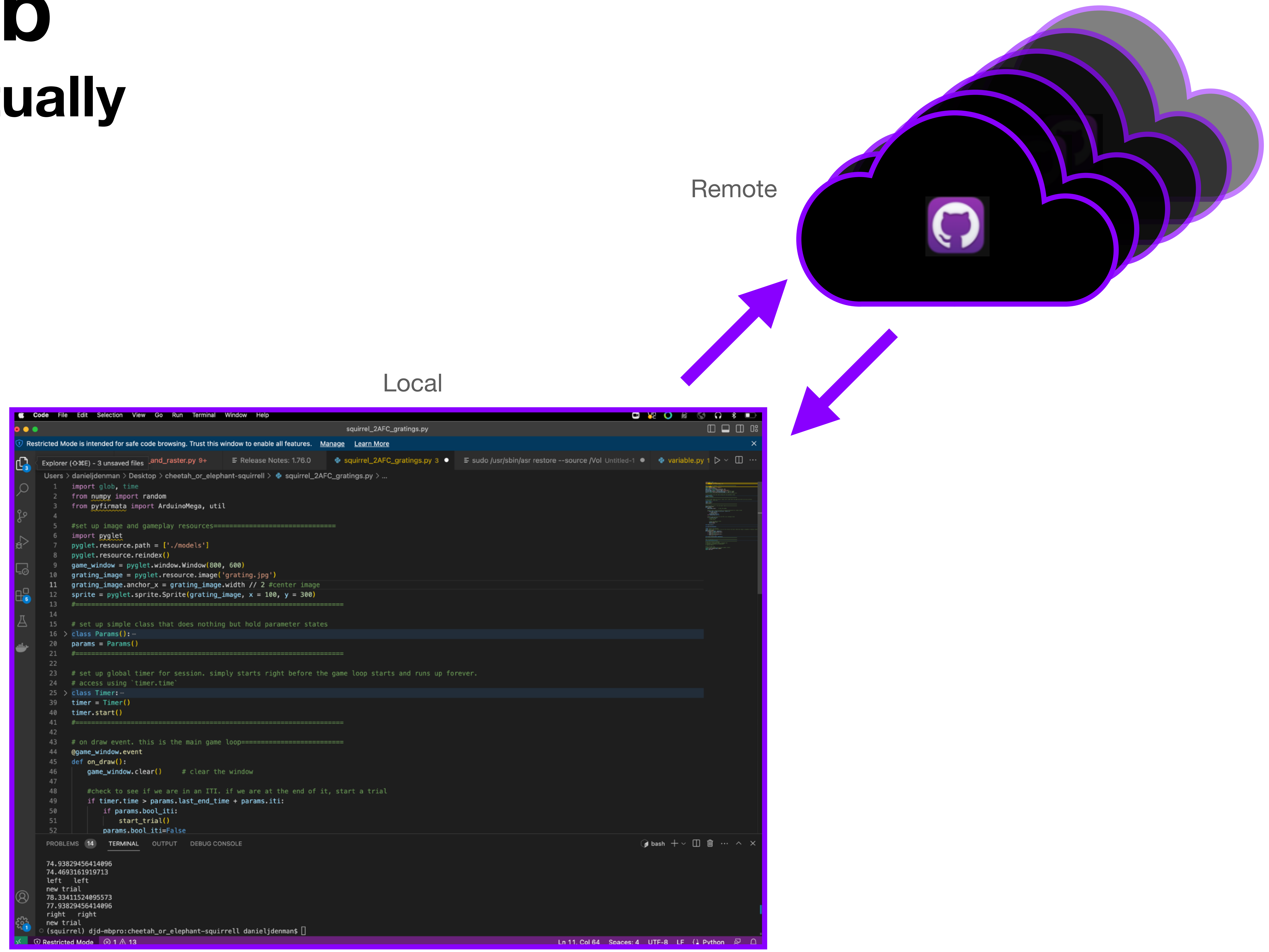

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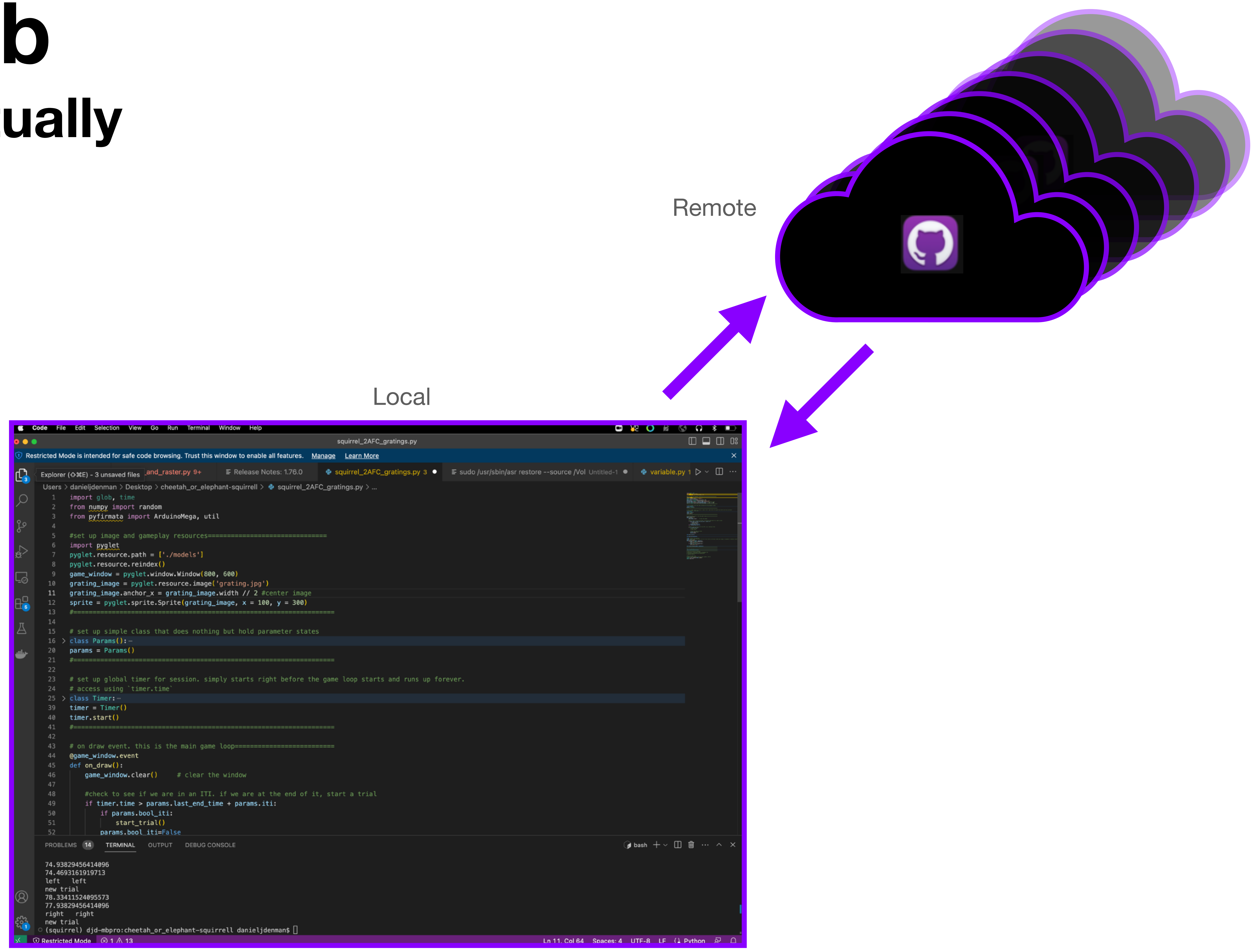

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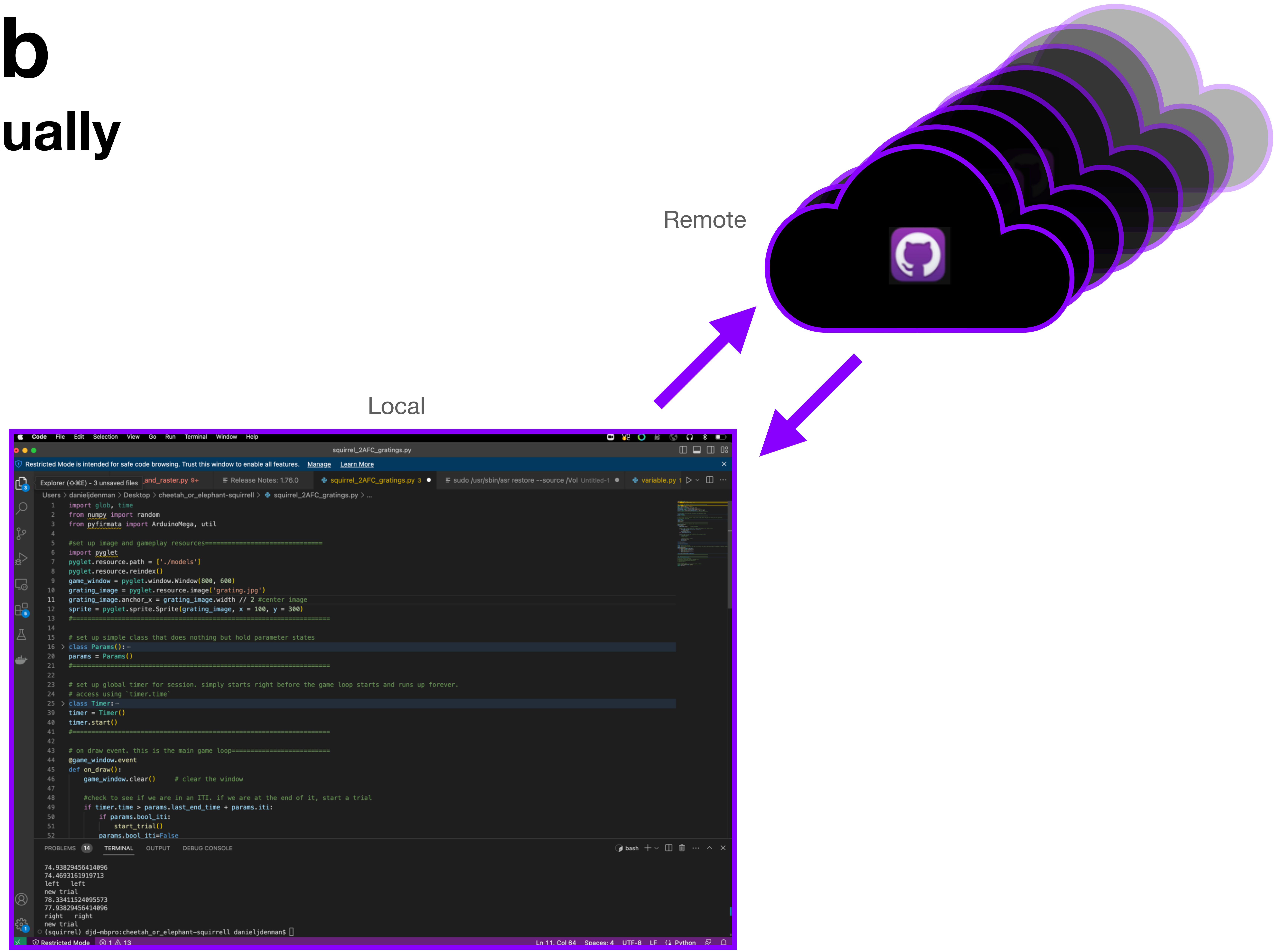

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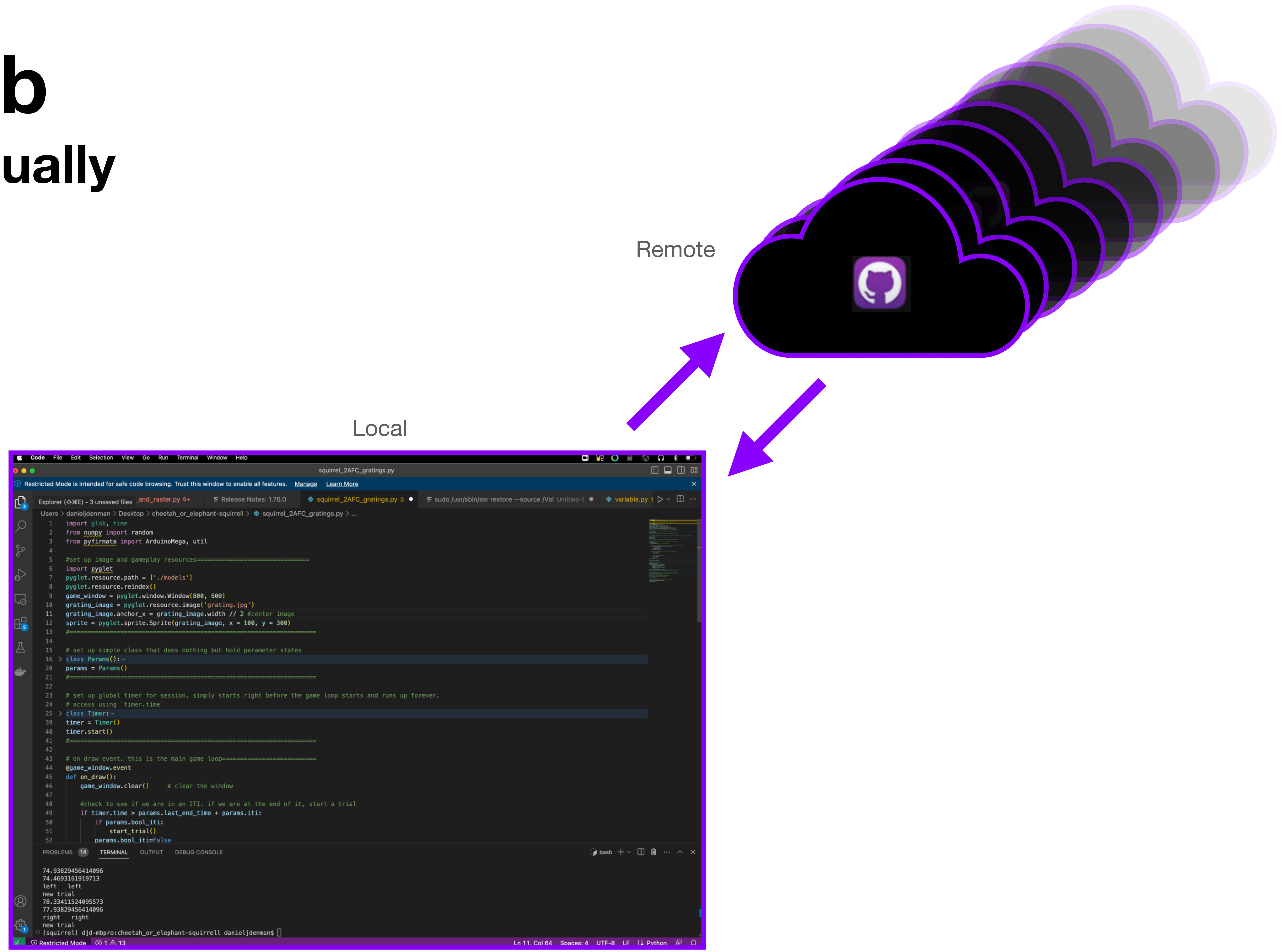

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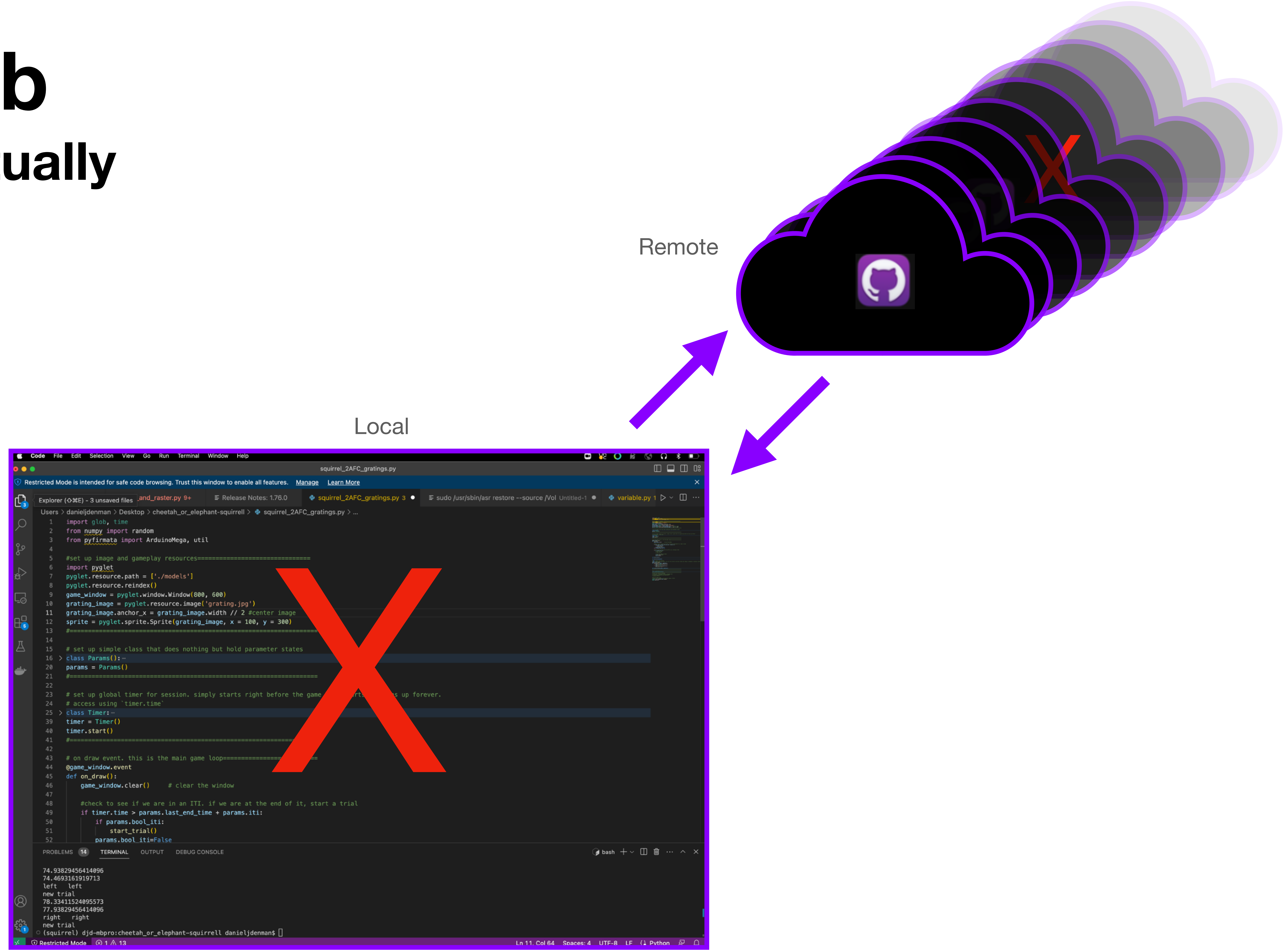

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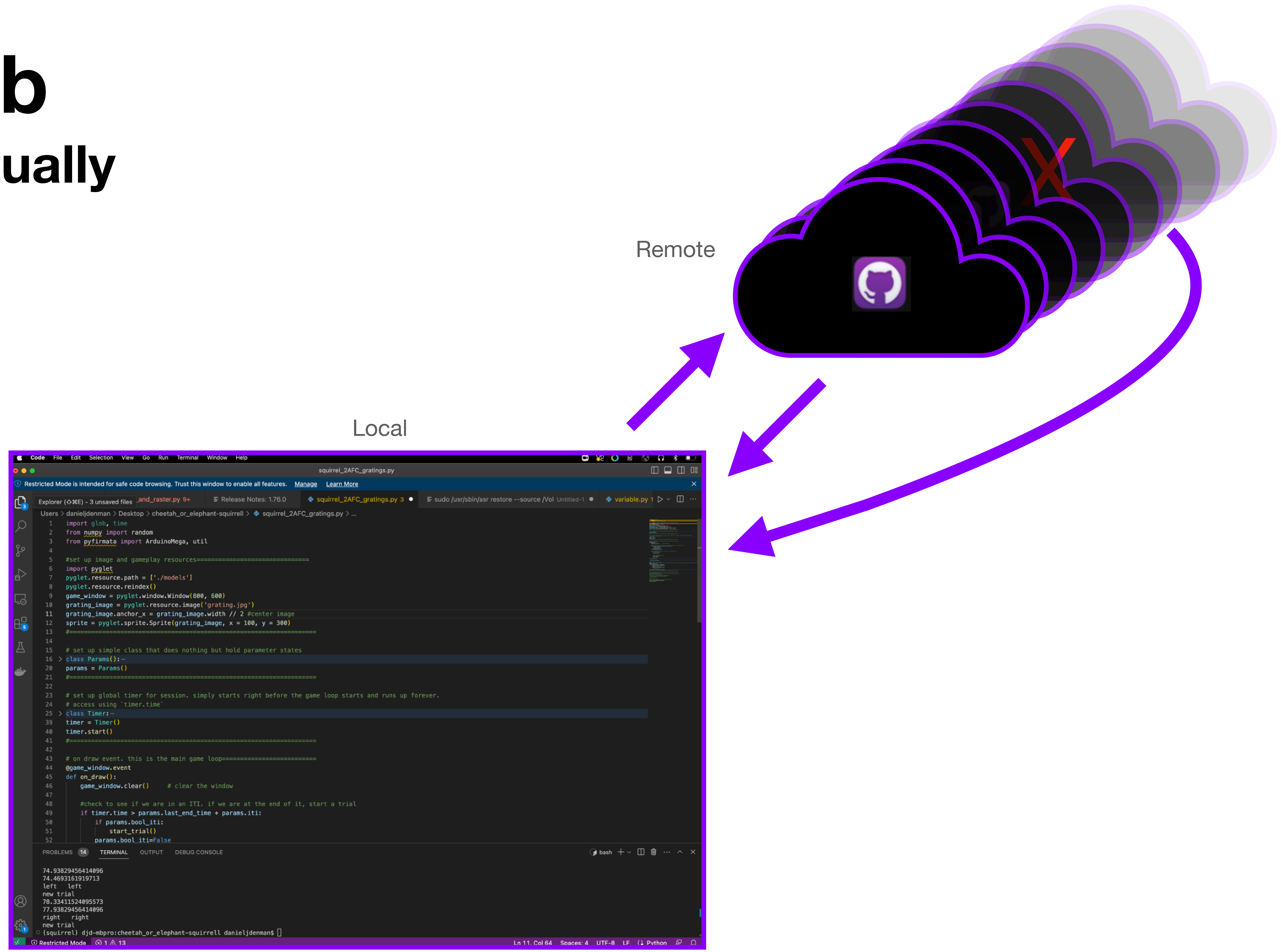
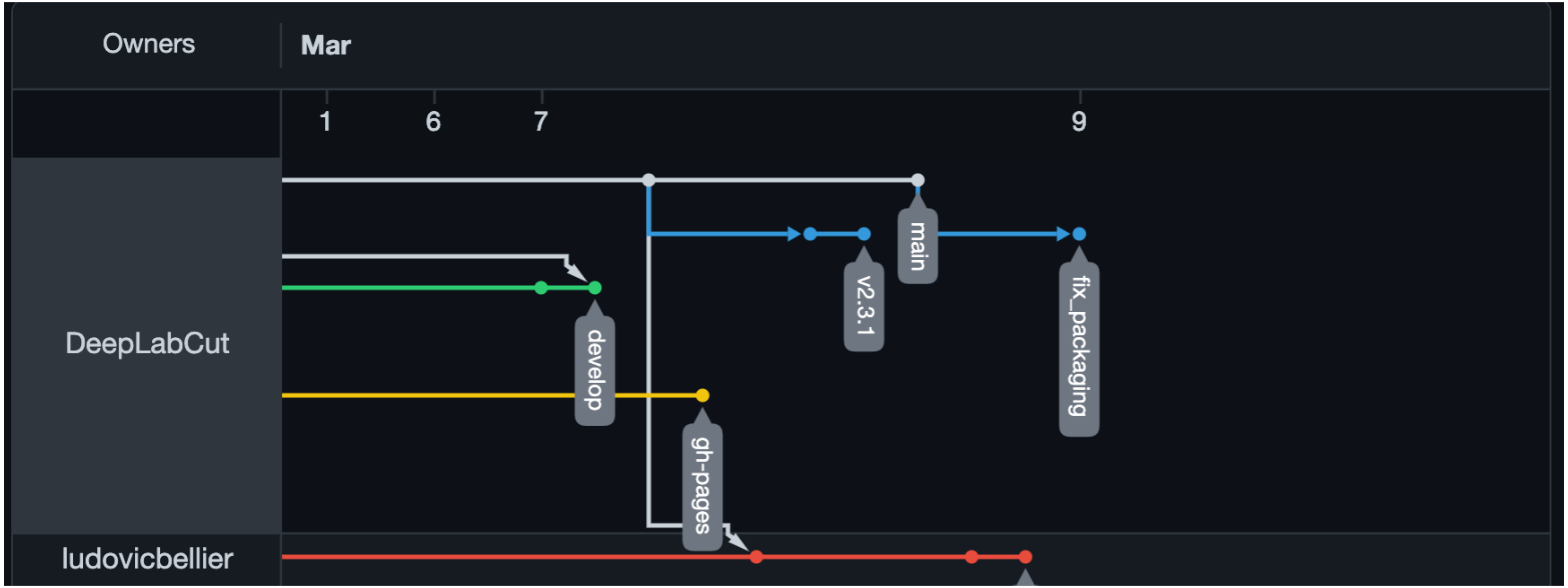


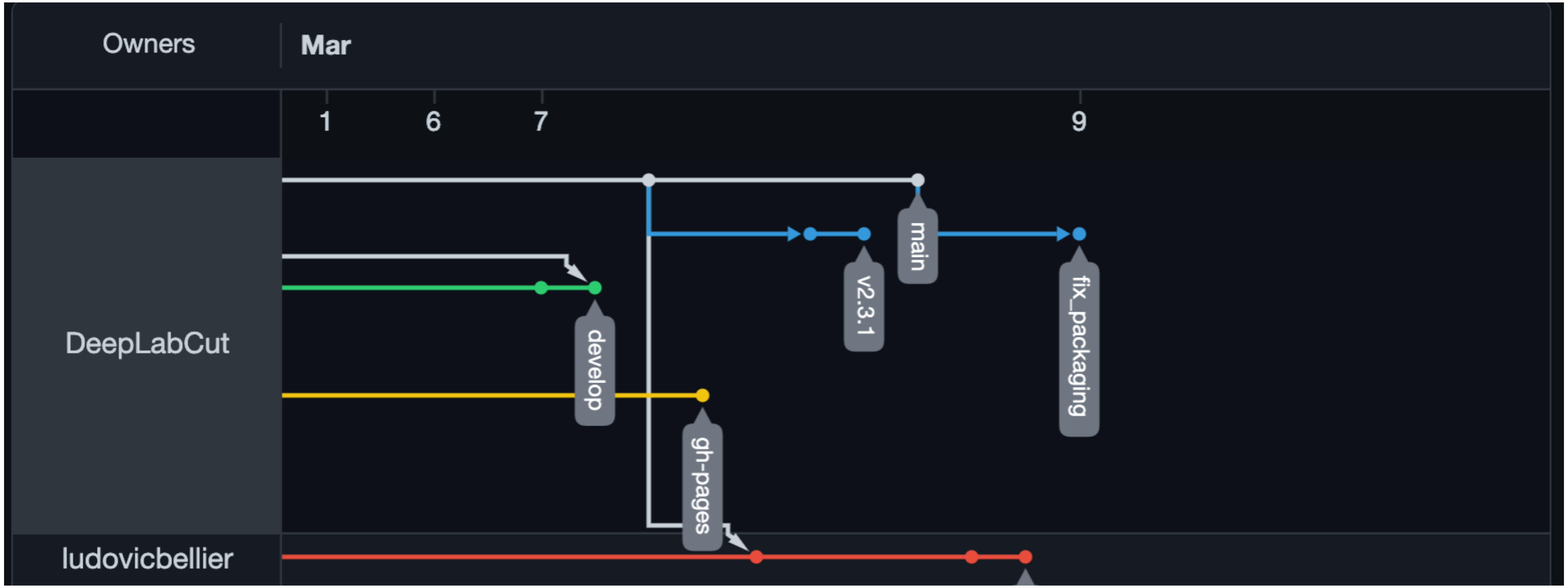
Github Desktop

Branching / Forking



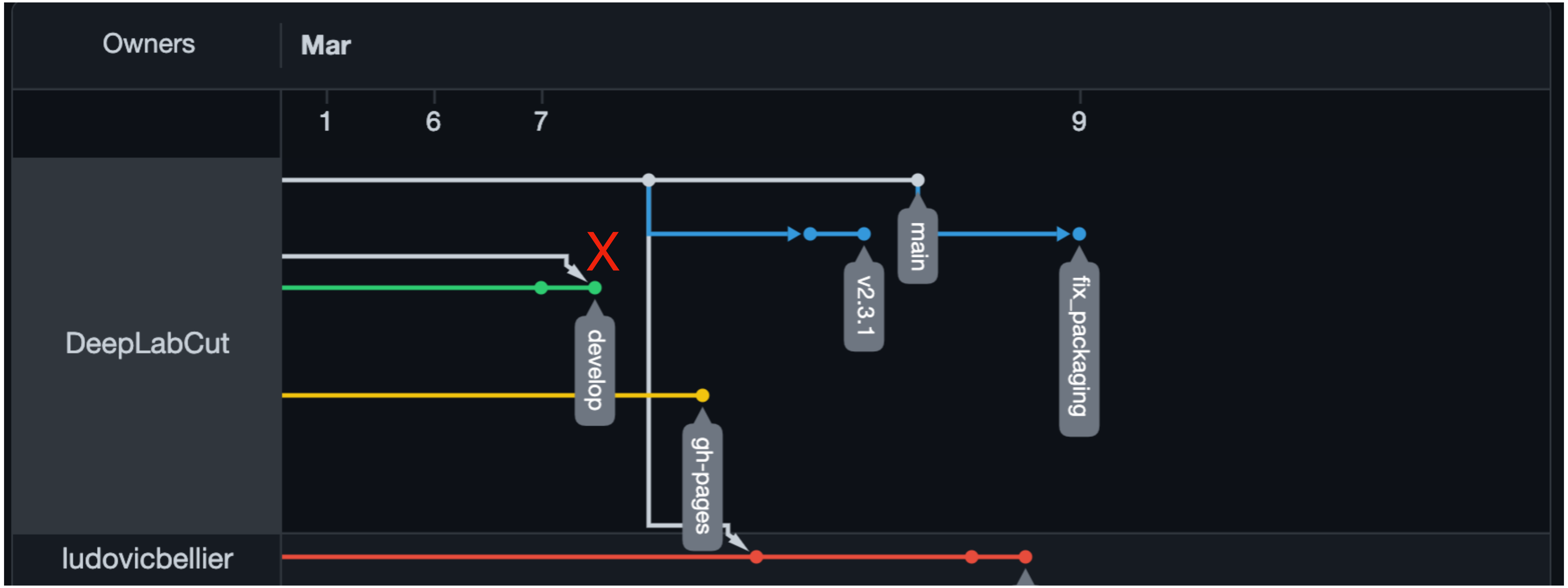
Github Desktop

Merging



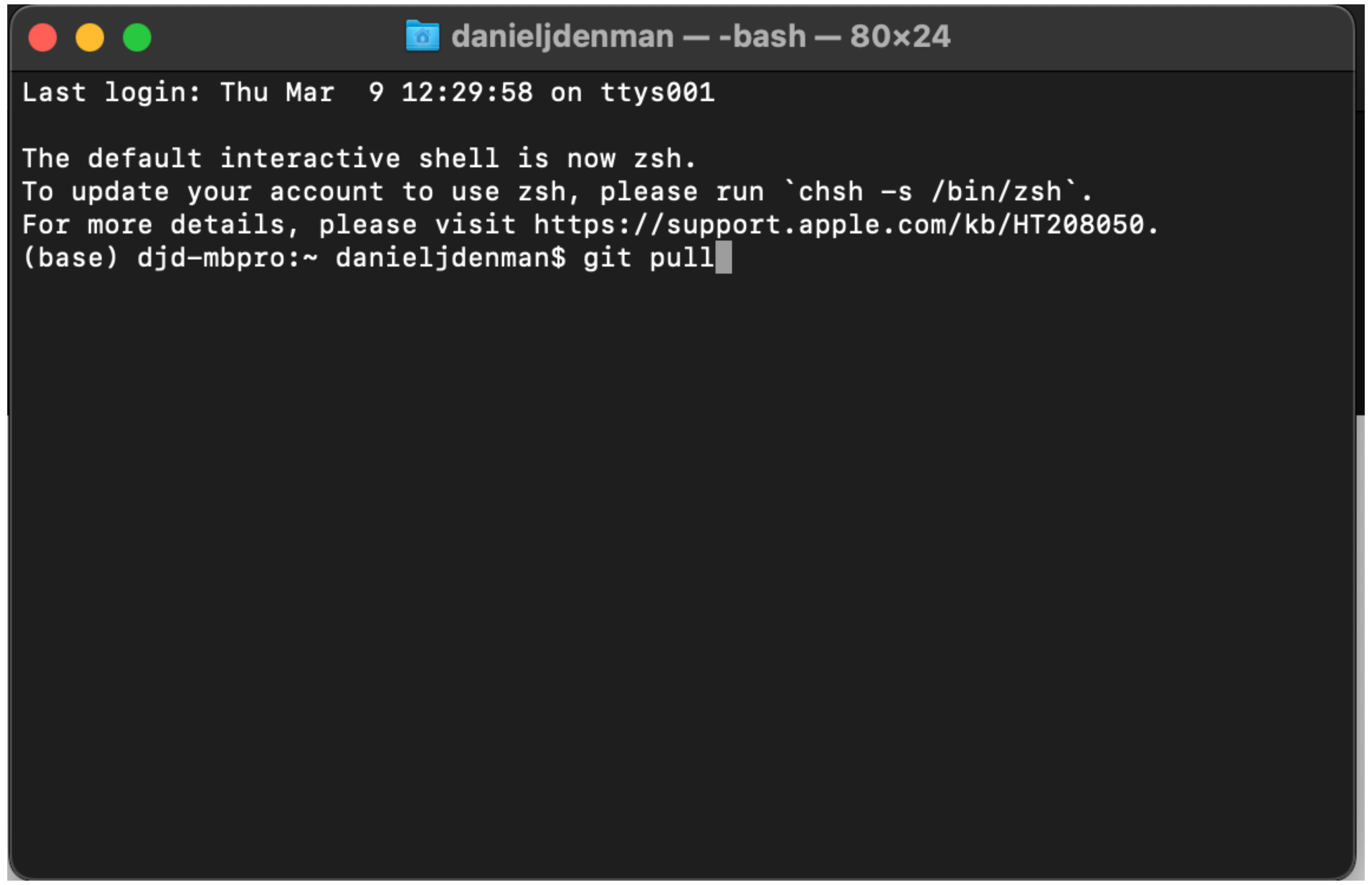
Github Desktop

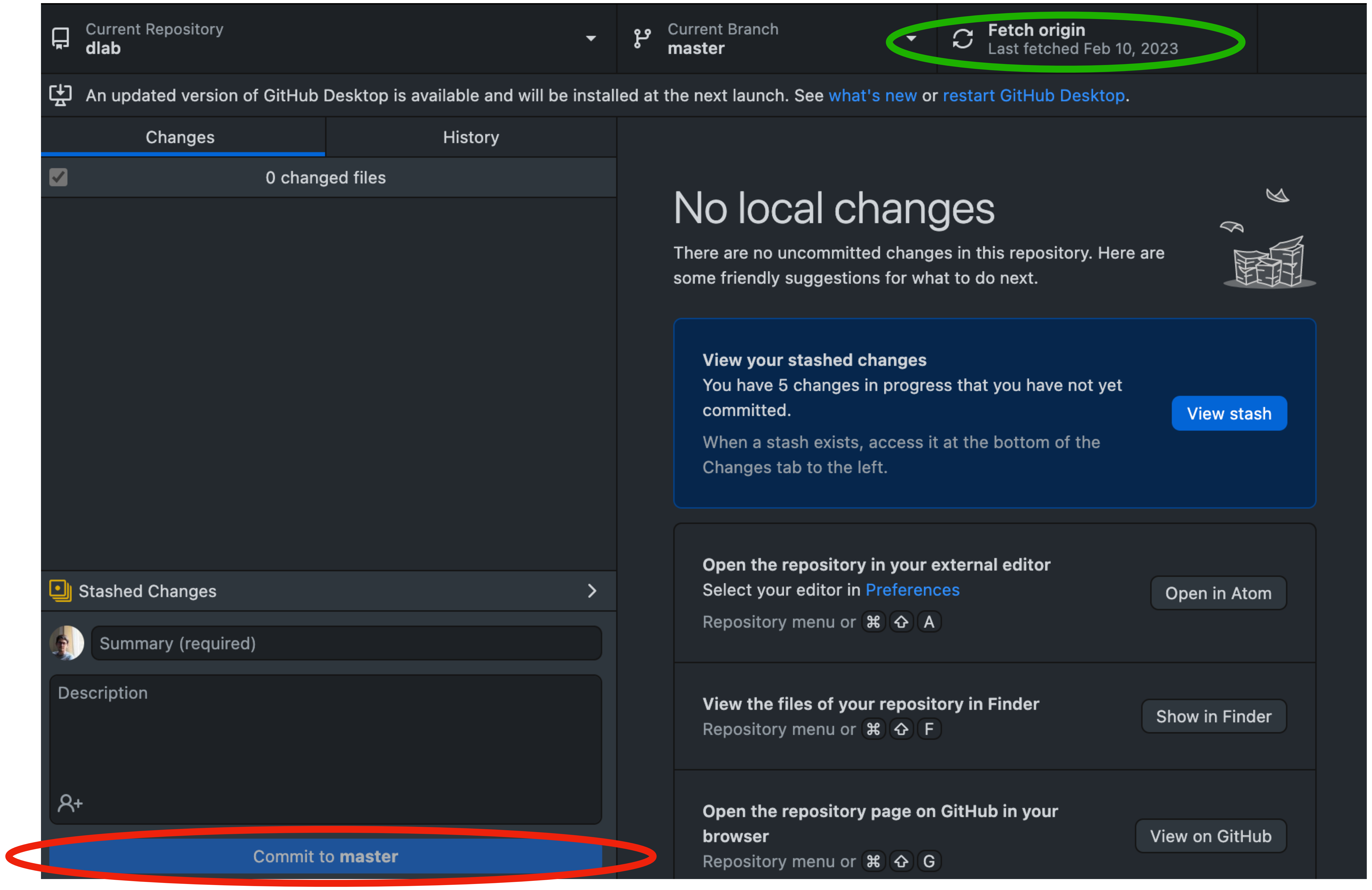
Conflicts

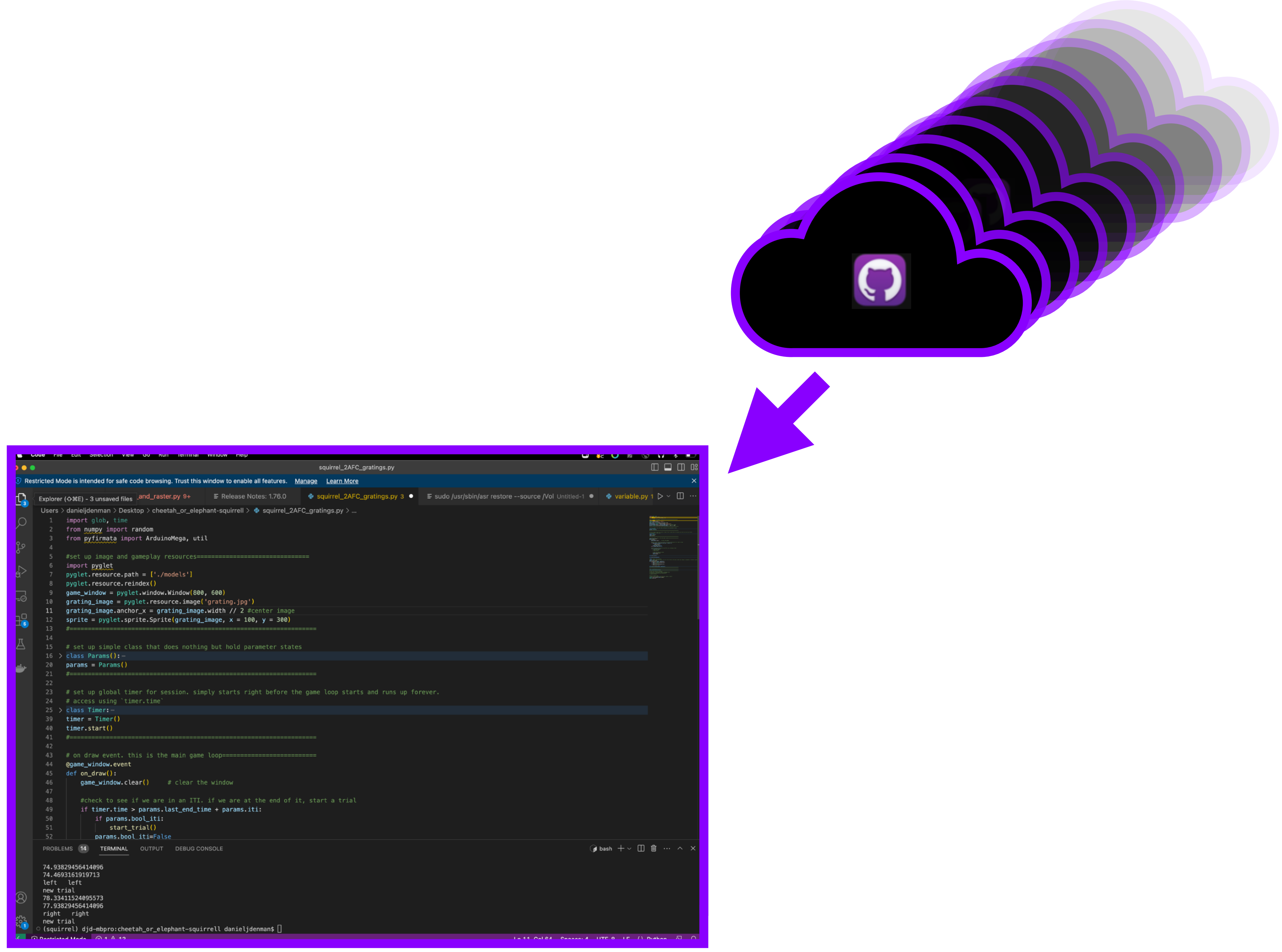


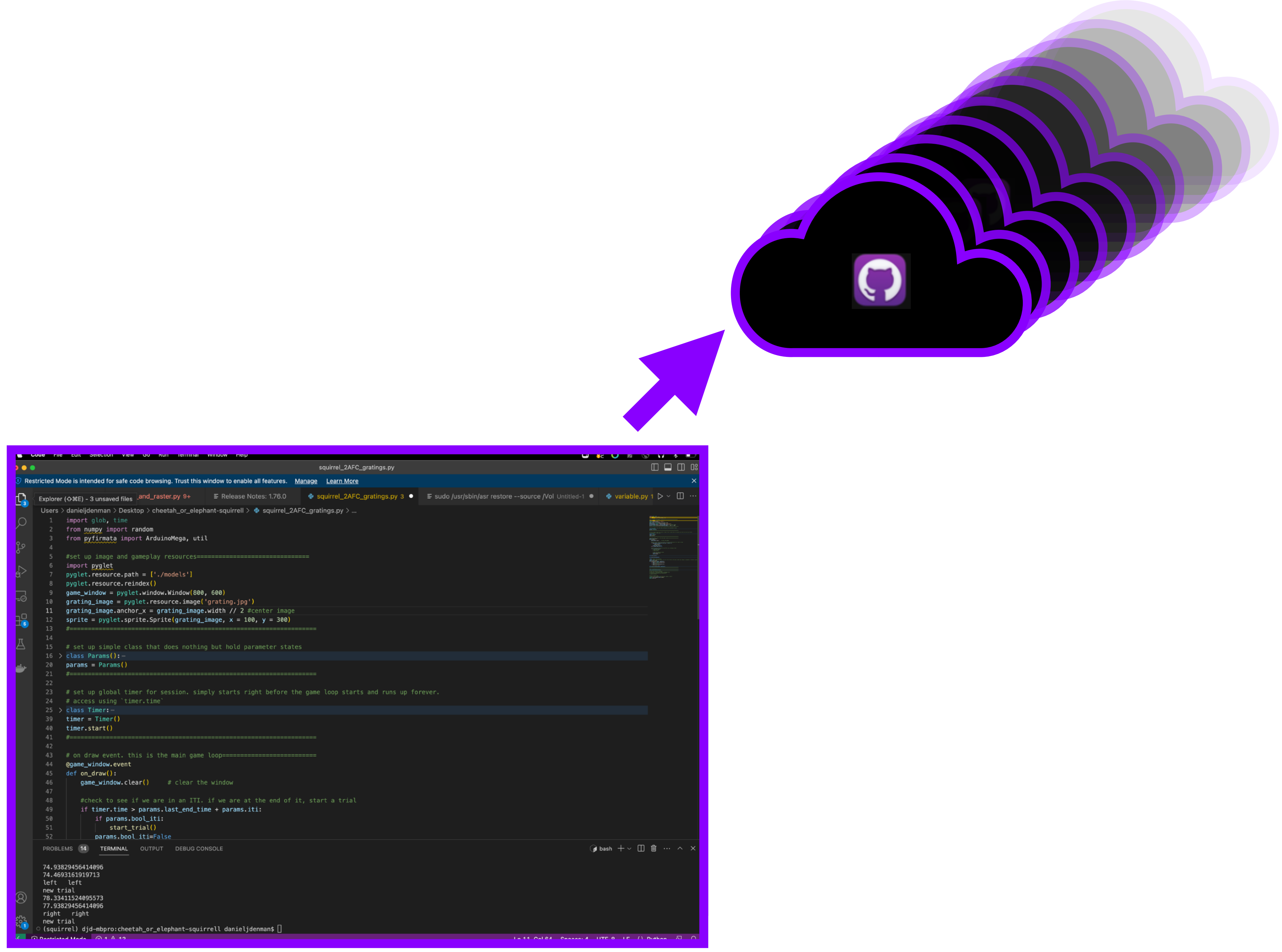
GitHub

CLI vs. GUI









```
git fetch origin
git reset --hard origin/master
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